

FULL STACK



Microsoft Certified Azure Administrator: AZ-104

FULL STACK

Manage Azure Identities and Governance



Learning Objectives

By the end of this lesson, you will be able to:

- Describe Azure's primary tools and services, and implement best practices for resource management and cost optimization
- Develop Azure groups, policies, RBAC, and resource locks, enhancing governance and security to minimize risks
- Apply and audit resources with Azure Policy to optimize Azure's identity services for enhanced security
- Use Microsoft Entra ID and Administer Identity to ensure secure and streamlined access to Azure resources



A Day in the Life of an Azure Administrator

You are working for an organization as an Azure Administrator. As your company plans to migrate to Azure, you need to determine how to acquire a subscription and how to effectively manage the cost of using Azure Services.

After obtaining the subscription, you need to decide who can access what types of Azure resources, so you need to devise an authentication strategy for Azure.

Once your access is set, you need to decide how to manage the infrastructure using the variety of tools available.

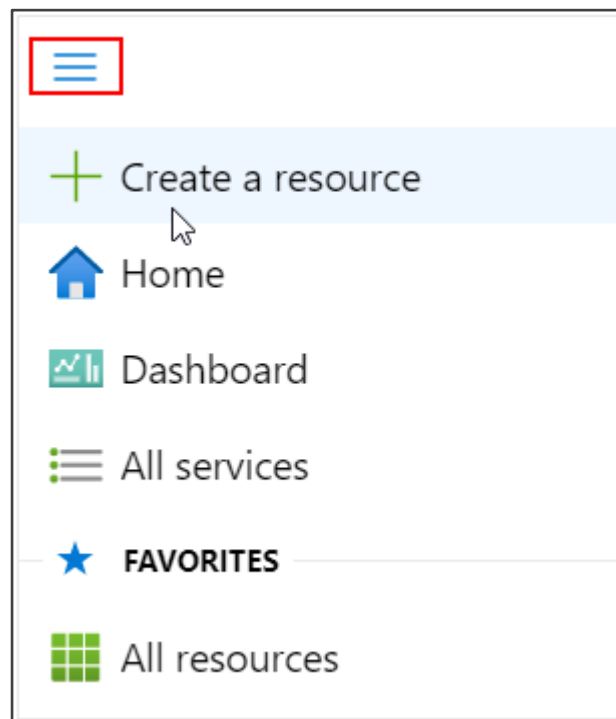


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Manage Subscriptions and Governance

Use the Azure Portal

It allows users to build, manage, and monitor everything from simple web apps to complex cloud applications in a single unified console.



The user can:

- Manage resources
- Create customized dashboards and reports
- Access cloud shell
- Receive notifications
- Search resources, services, and documentation

Source: <https://learn.microsoft.com/en-us/azure/azure-portal/azure-portal-overview>

Creating a Microsoft Azure Account



Duration: 10 min.

Problem Statement:

You've been tasked with creating an Azure free account and taking a quick tour of the Azure portal.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Go to <https://azure.microsoft.com/free>
2. Create an Azure free account
3. Create a resource group



Use Azure Cloud Shell

It is used to manage Azure resources, which provides the flexibility of choosing the shell experience that best suits the way one works.



Is an interactive,
browser-
accessible shell

Offers either Bash
or PowerShell

Authenticates
automatically

Walkthrough of Azure Cloud Shell



Problem Statement:

You've been tasked with providing a walkthrough of Azure cloud shell.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Sign-in to the Azure portal
2. Launch Cloud Shell from the Azure portal
3. Navigate to your choice of Azure command-line
4. Run commands based on the selected environment



Use Azure PowerShell

Features

- Helps connect to one's Azure subscription and manage resources
- Adds Azure-specific commands
- Is available inside a browser via the Azure Cloud Shell or inside a Docker container
- Is available as a local installation on Linux, macOS, and Windows

Syntax

```
New-AzVm `
  -ResourceGroupName "myResourceGroup" `
  -Name "myVM" `
  -Image "UbuntuLTS" `
  ...
```


Use Azure CLI

It is a command-line interface to connect to Azure and execute administrative commands on Azure resources.

Command

```
az vm restart -g MyResourceGroup -n MyVm
```

Azure CLI

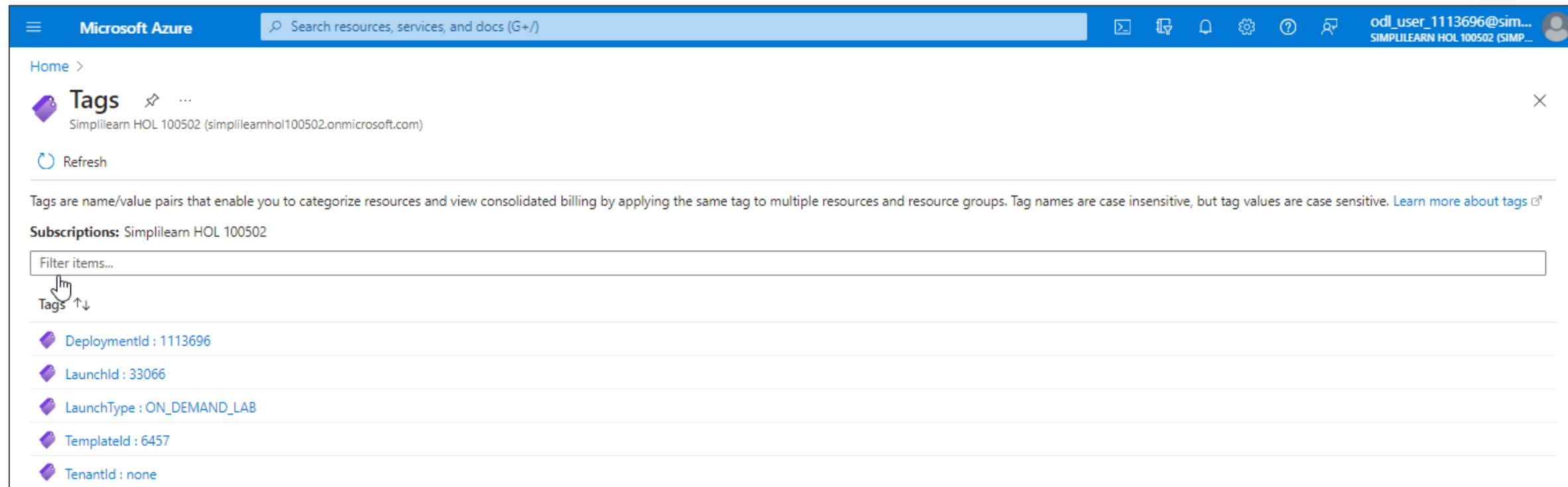
- Is a cross-platform command-line program
- Runs on Linux, macOS, and Windows
- Can be used interactively or through scripts



Azure Tags

Tags consist of a name-value pair. Applying tags to the Azure resources serves the following purposes:

- Provides metadata for the Azure resources
- Logically organizes resources into a taxonomy
- Enables grouping of billing data



The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Tags' page for the subscription 'Simplilearn HOL 100502'. It includes a 'Refresh' button, a description of tags, and a list of existing tags. A 'Filter items...' search bar is also present.

Microsoft Azure

Search resources, services, and docs (G+)

Home >

Tags

Simplilearn HOL 100502 (simplilearnhol100502.onmicrosoft.com)

Refresh

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags](#)

Subscriptions: Simplilearn HOL 100502

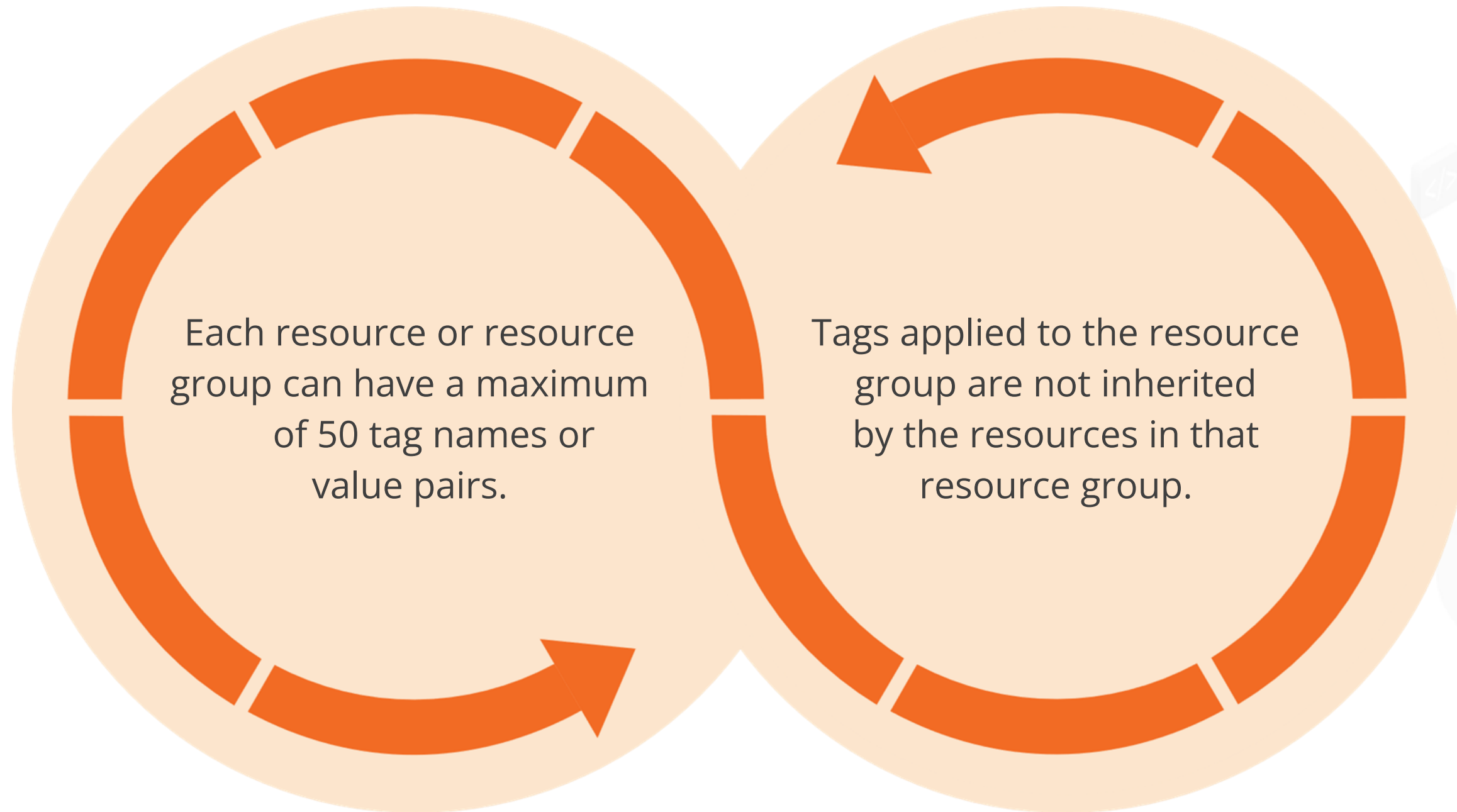
Filter items...

Tags ↑↓

- DeploymentId : 1113696
- LaunchId : 33066
- LaunchType : ON_DEMAND_LAB
- TemplateId : 6457
- TenantId : none

Azure Tags

Points to remember for using Azure Tags:



Creating and Assigning Tags to Resources via Azure Portal



Problem Statement:

You've been tasked with creating and assigning tags to resources via the Azure portal.

ASSISTED PRACTICE

Assisted Practice: Guidelines

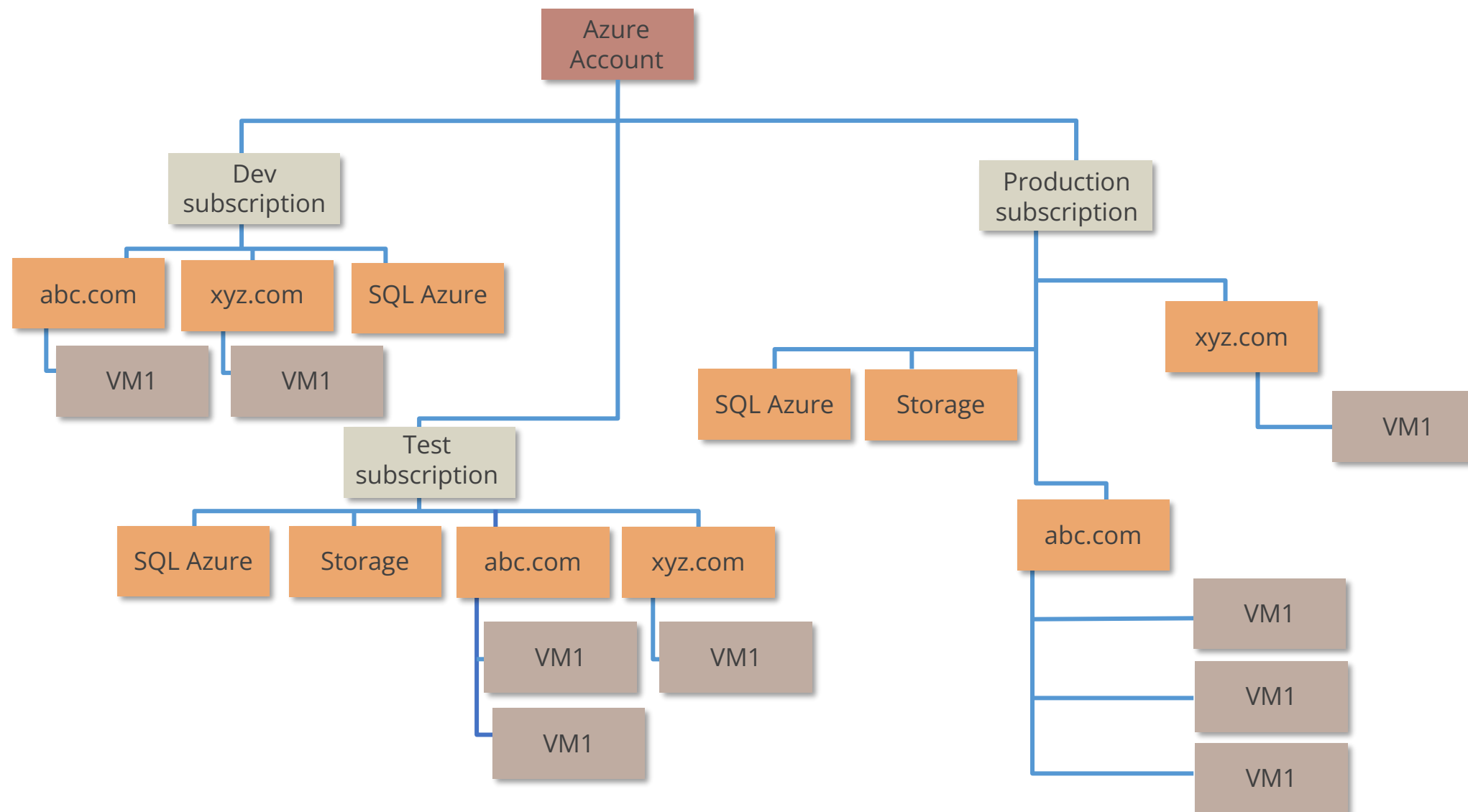
Steps to be followed:

1. Create Azure tags



Implement Azure Subscriptions

- It is a unit of Azure services that is linked to an Azure account.
- Azure services are billed on a per-subscription basis



Source: <https://learn.microsoft.com/en-us/archive/blogs/arunrakwal/create-windows-azure-subscription>

Obtain a Subscription

Following are the ways to obtain an Azure subscription:

- Enterprise Agreement: Customers pay upfront and consume services throughout the year
- Resellers: Users can buy Azure through an open licensing program that provides a simple and flexible way to purchase cloud services
- Partners: Users can design and implement their Azure cloud solution
- Personal free account: Users can start right away



Enterprise



Resellers



Partners



Personal

Identify Subscription Usage

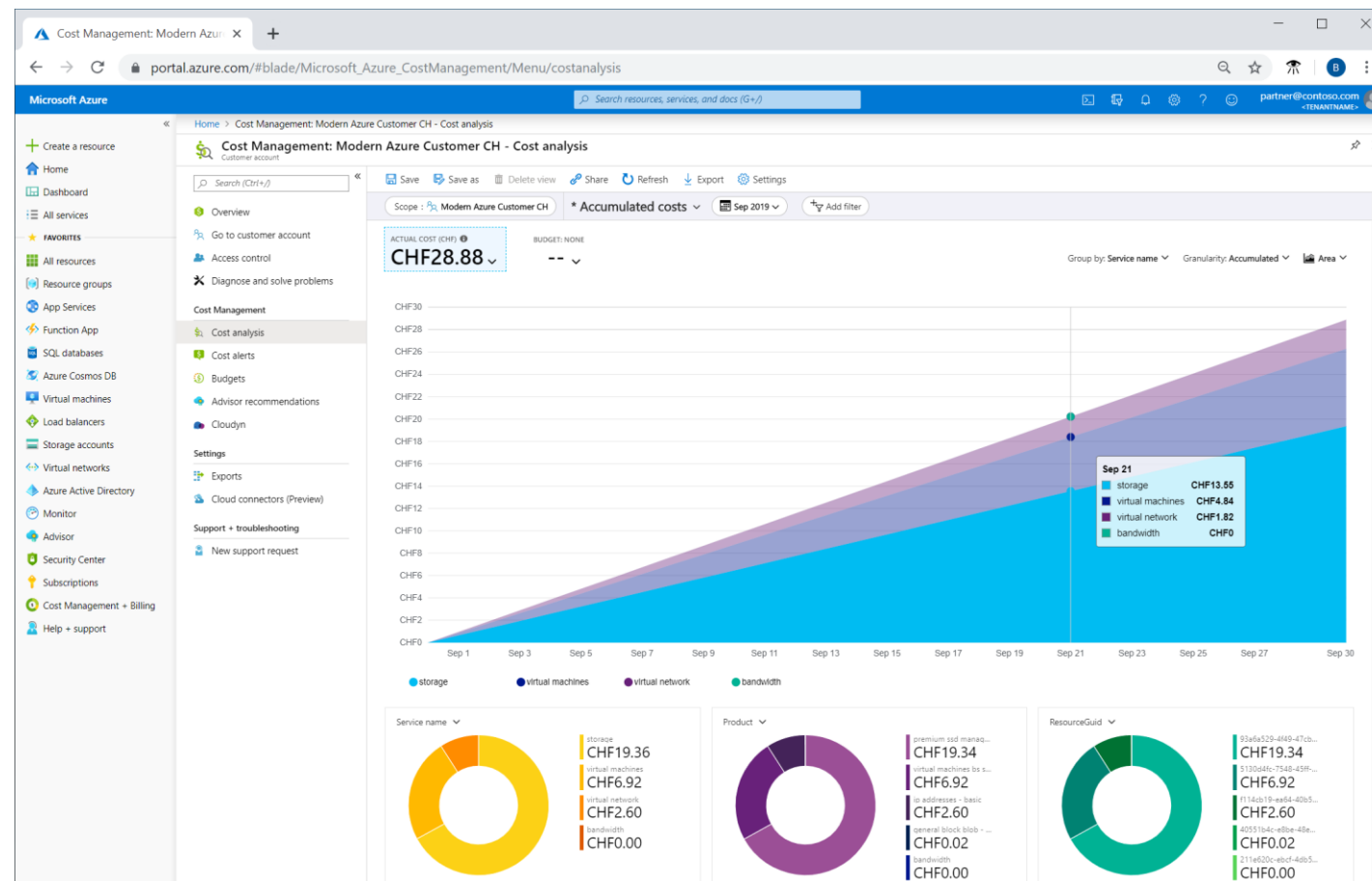
The table below depicts the different subscriptions and their usage:

Subscription	Usage
Free	Includes a \$200 credit for the first 30 days, free limited access for 12 months
Pay-As-You-Go	Charges monthly
CSP	Agreement with possible discounts through a Microsoft Cloud Solutions provider partner usually for small to medium businesses
Enterprise	A single agreement, with discounts for new licenses and Software Assurance, targeted at enterprise-scale organizations
Student	Includes \$100 for 12 months; must verify student access



Implement Cost Management

It displays organizational cost and usage patterns using advanced analytics.



Cost management helps to:

- Conduct cost analysis to explore and analyze organizational costs
- Create a budget that helps prevent cost thresholds or limits from surpassing
- Review recommendations to identify idle and underutilized resources
- Export the data used by the external systems to access or review cost management data

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Azure Advisor

Introduction to Azure Advisor

It is a personalized cloud consultant that helps users follow best practices for optimizing Azure deployment.



As a result of the analysis, it offers recommendations for Azure resources to improve cost-effectiveness, performance, reliability, and security.

Introduction to Azure Advisor

The advisor allows you to do the following:

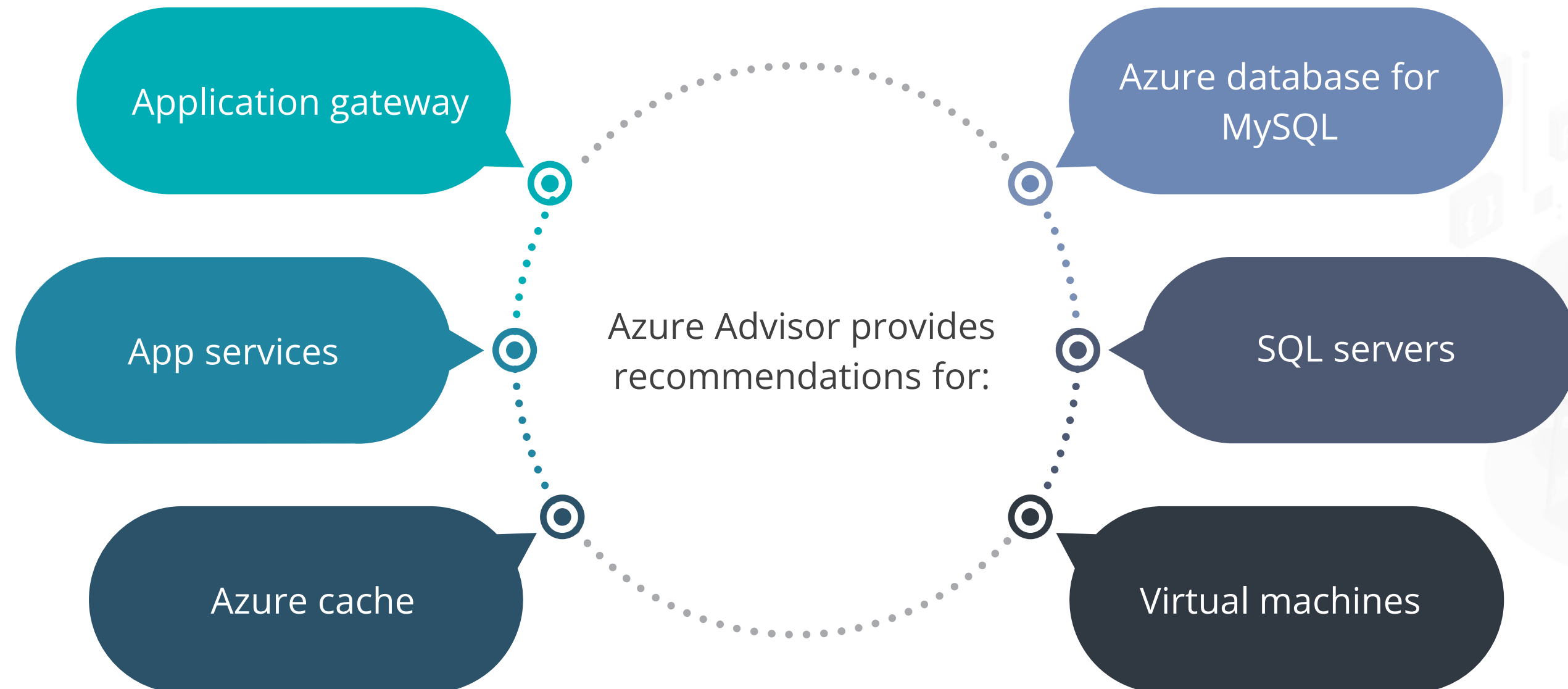
Discover proactive, actionable, and personalized best practice recommendations

Improve resource performance, security, and reliability, and find ways to reduce Azure resource costs

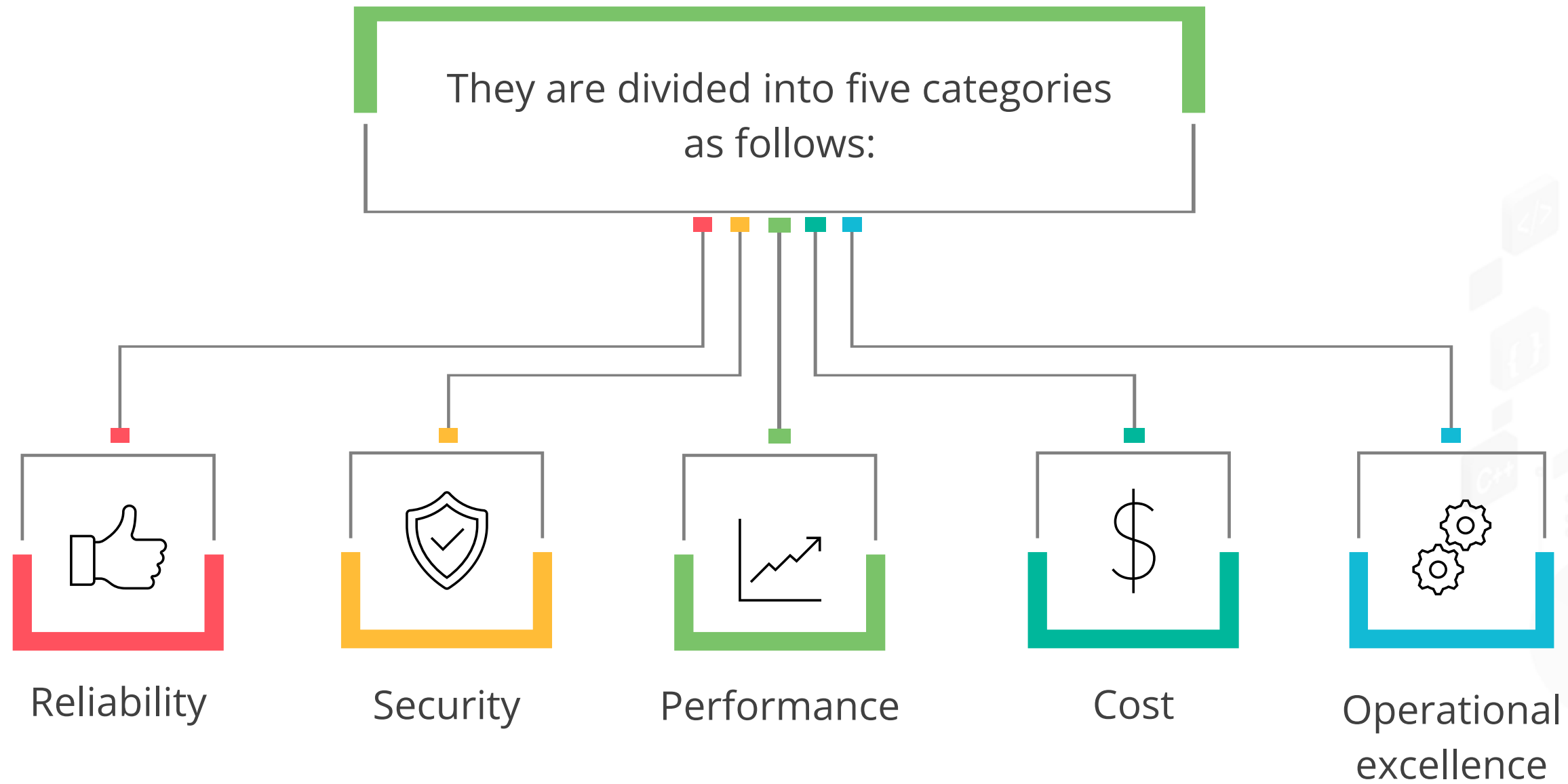
Access recommendations with inline proposed actions

Azure Advisor Recommendations

Access is given to an owner, contributor, reader, resource group, or resource.



Azure Advisor Recommendations



Advisor Reliability Recommendations

It can be found on the reliability tab of the Azure advisor. It helps ensure and improve the continuity of the business-critical applications.

Here's how the Azure Advisor can be used to improve the reliability of user's application:

Check the version of your check point network virtual appliance image

Ensure application gateway fault tolerance

Protect VM data from accidental deletion

Access Azure experts as needed

Advisor Security Recommendations

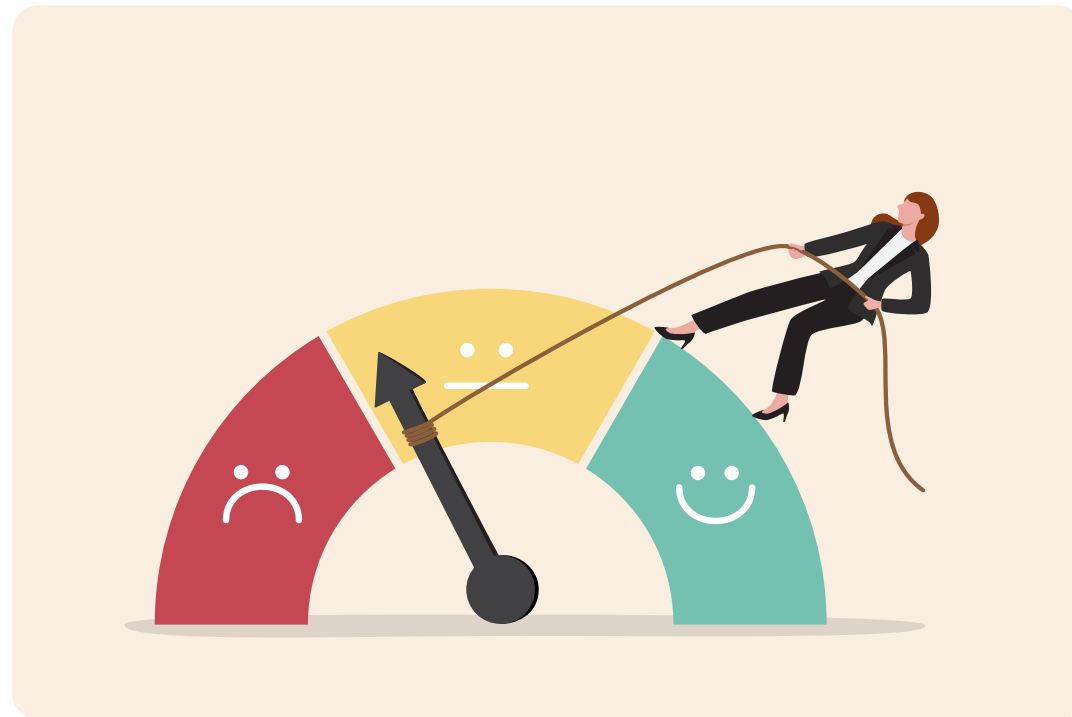
It can be viewed on the security tab on the Advisor dashboard. Azure Advisor integrates with Microsoft Defender for the cloud to provide security recommendations.



- Defender for Cloud enhances visibility and control over Azure resource security.
- It helps in preventing, detecting, and responding to threats.

Advisor Performance Recommendations

It can be viewed on the performance tab on Azure Advisor.

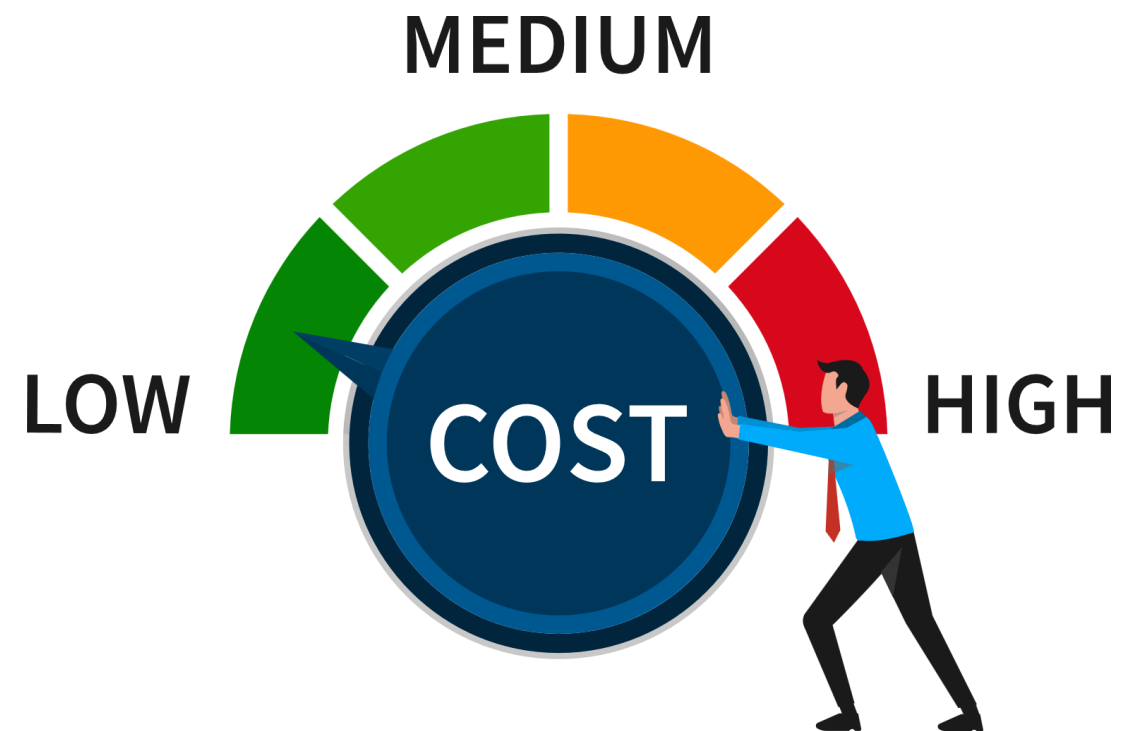


It helps to:

- Improve the speed and responsiveness of the business-critical applications
- Shorten DNS time-to-live (TTL) in the traffic manager profile for faster failover to healthy endpoints
- Use SQL database advisor to improve database performance
- Update storage client library for better performance and reliability
- Enhance app service performance and reliability

Advisor Cost Recommendations

It can be viewed on the cost tab on Azure Advisor.



This helps to:

- Optimize spending on VMs or virtual machine scale sets (VMSS) by resizing or shutting down instances that are underutilized.
- Shutdown recommendations for the resources that have remained unused for the past 7 days.
- Resize SKU recommendations
- Give burstable recommendations

Advisor Operational Excellence Recommendations

It can be viewed on the operational excellence tab on Azure Advisor.



This helps in:

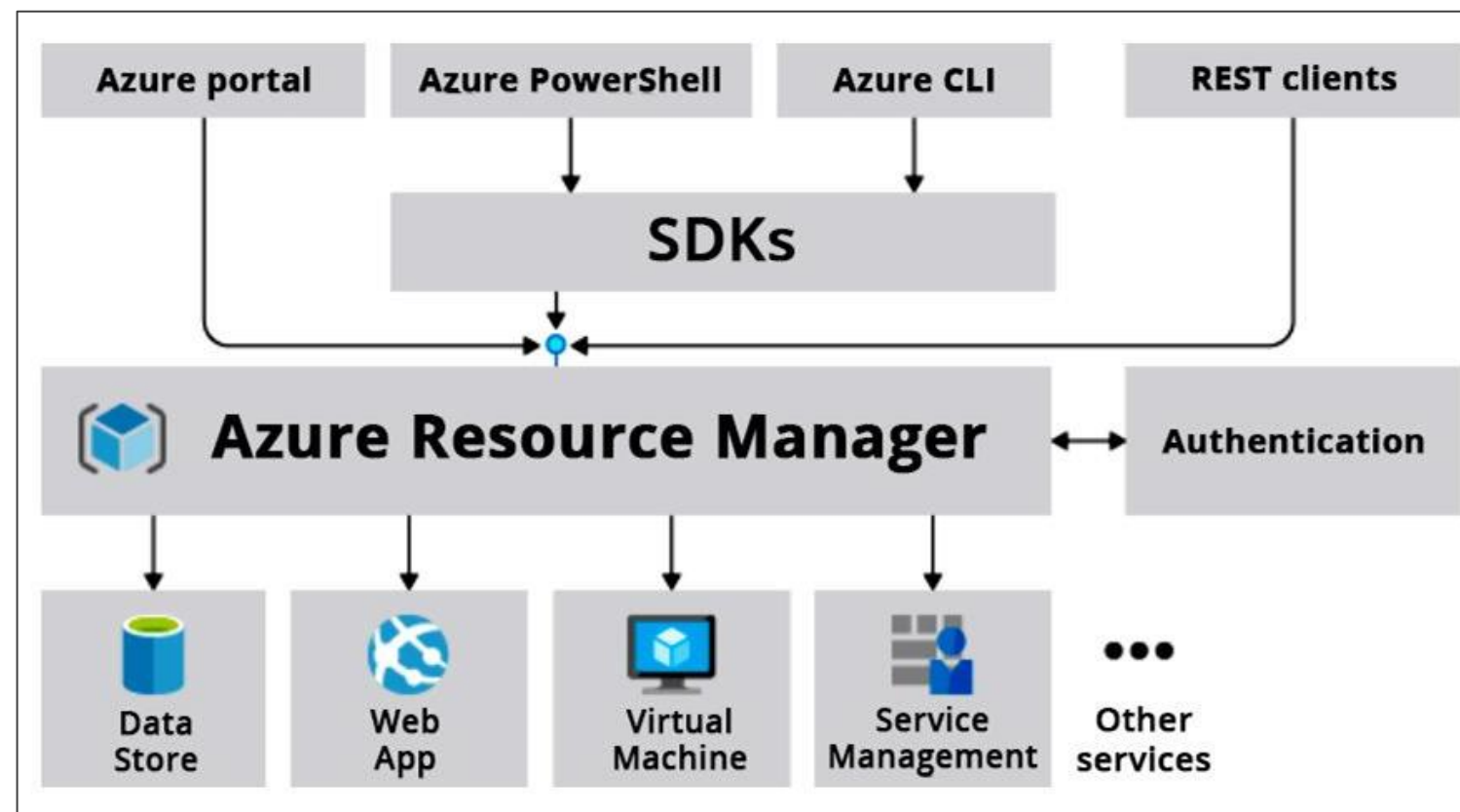
- Process and workflow efficiency
- Resource manageability
- Deployment best practice

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Azure Resource Manager

Resource Manager

It is Azure's deployment and management service that enables the user to create, update, and delete resources in their Azure account.



Benefits

- Provides a consistent management layer
- Enables the user to work with the resources in a solution as a group
- Allows the user to deploy, update, or delete in a single, coordinated operation
- Provides security, auditing, and tagging features

Azure Resource Terminology

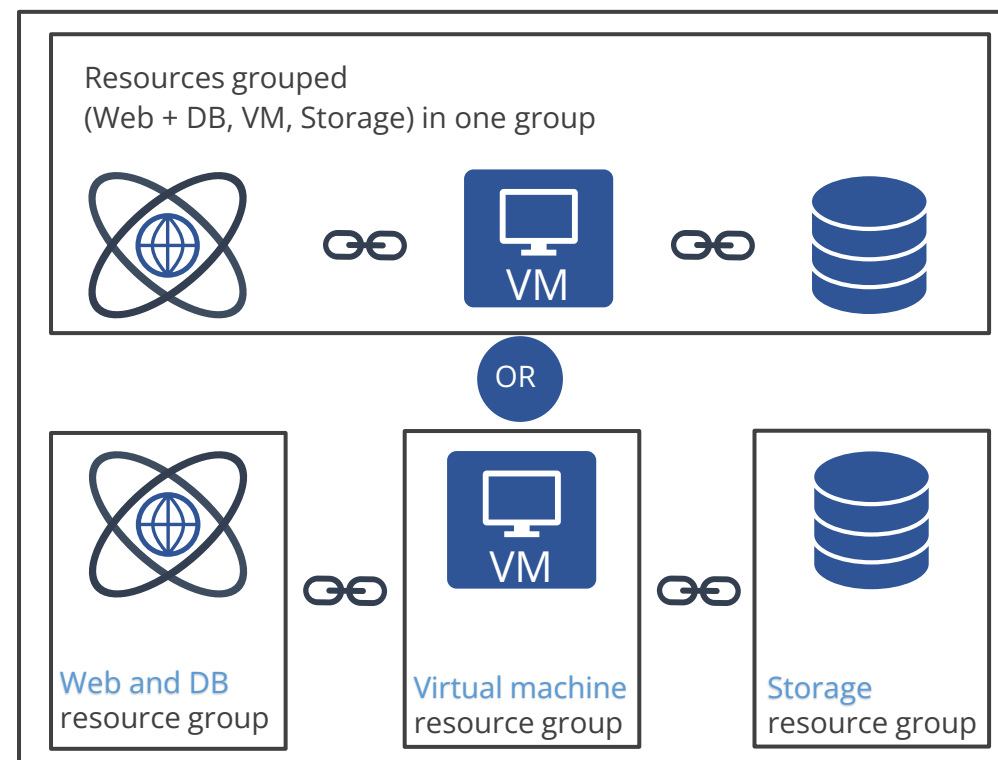
- Resource: It is a single service instance in Azure, such as Azure Virtual Machines, virtual networks, and storage accounts.
- Subscription: It is a logical container for one's resources. Each Azure resource is associated with only one subscription.
- Resource group: It is a logical container that the user uses to group related resources in a subscription. Each resource can exist in only one resource group.
- Management groups: It refers to the Logical containers that one uses for one or more subscriptions.
- Azure account: It is the email address that the user provides when they create an Azure subscription.

Azure Resource Terminology

- Region: It is a set of Azure datacenters deployed inside a latency-defined perimeter. The datacenters are connected through a dedicated, regional, low-latency network.
- Microsoft Entra ID(Azure AD): It is a Microsoft cloud-based identity and access management service. It allows a user's employees to sign in and access resources.
- Microsoft Entra tenant: It is a dedicated and trusted instance of Microsoft Entra ID. It is automatically created when an organization signs up for a Microsoft cloud service subscription. It represents a single organization.
- Microsoft Entra directory: Each Microsoft Entra tenant has a single, dedicated, and trusted directory. The directory includes the tenant's users, groups, and applications.

Resource Groups

It is a container that holds related resources for an Azure solution.

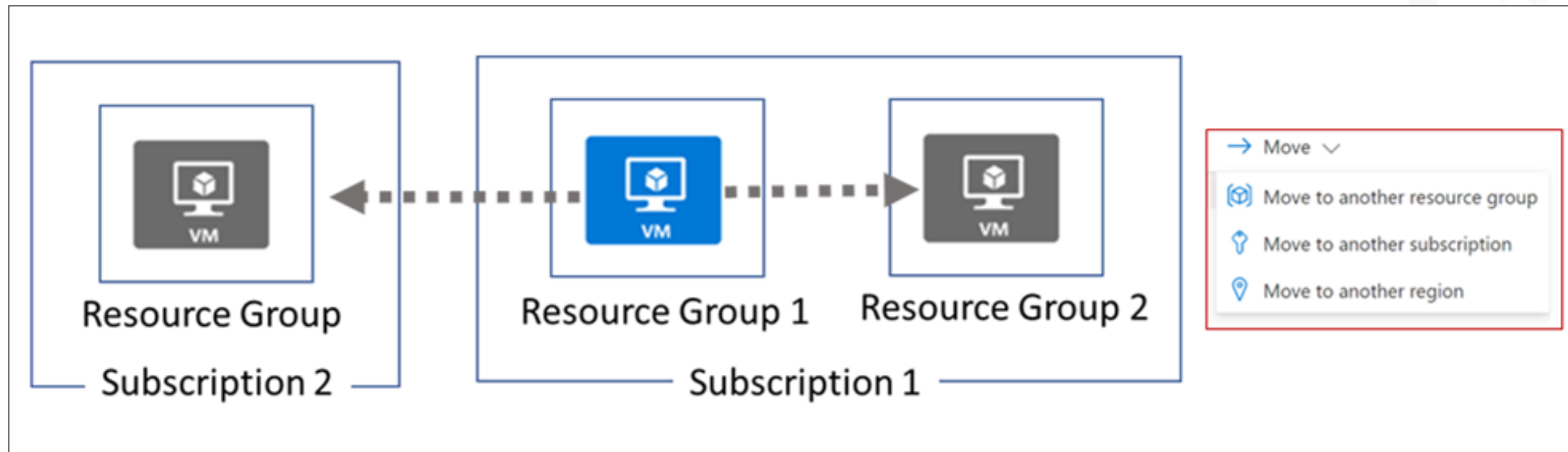


Features

- Resources can only exist in one resource group.
- Groups can have resources of many different types (services) and from many different regions.
- Groups cannot be renamed or nested.

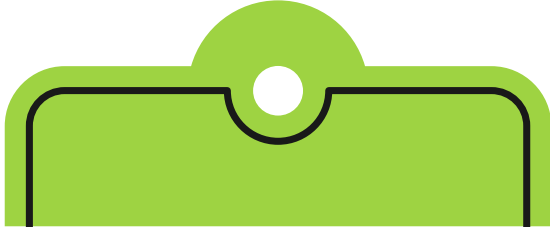
Reorganize Azure Resources

- When moving resources, the source group and the target group are locked.
- The services that cannot be moved are Microsoft Entra ID Domain Services, ExpressRoute, and Site Recovery.

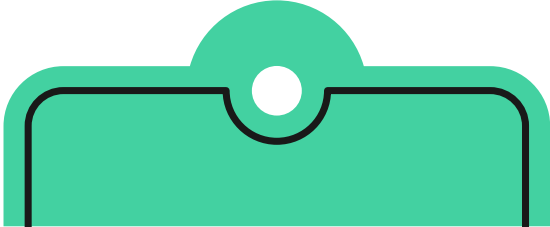


Determine Resource Limits

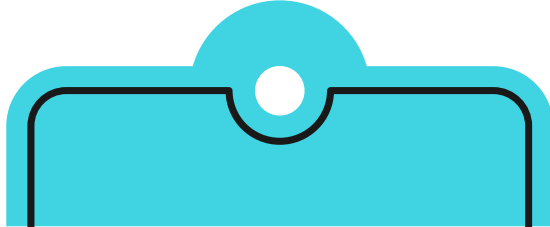
Azure lets the user view resource usage against limits.



Resources have a default limit or quota.



This is helpful to track current usage and plan for future use.



A free support case increases limits to published maximums.



Determine Resource Limits

The following image depicts the way of determining resource limits:

ASC DEMO | Usage + quotas
Subscription

Settings

- Programmatic deployment
- Resource groups
- Resources
- Usage + quotas**
- Policies
- Security
- Events

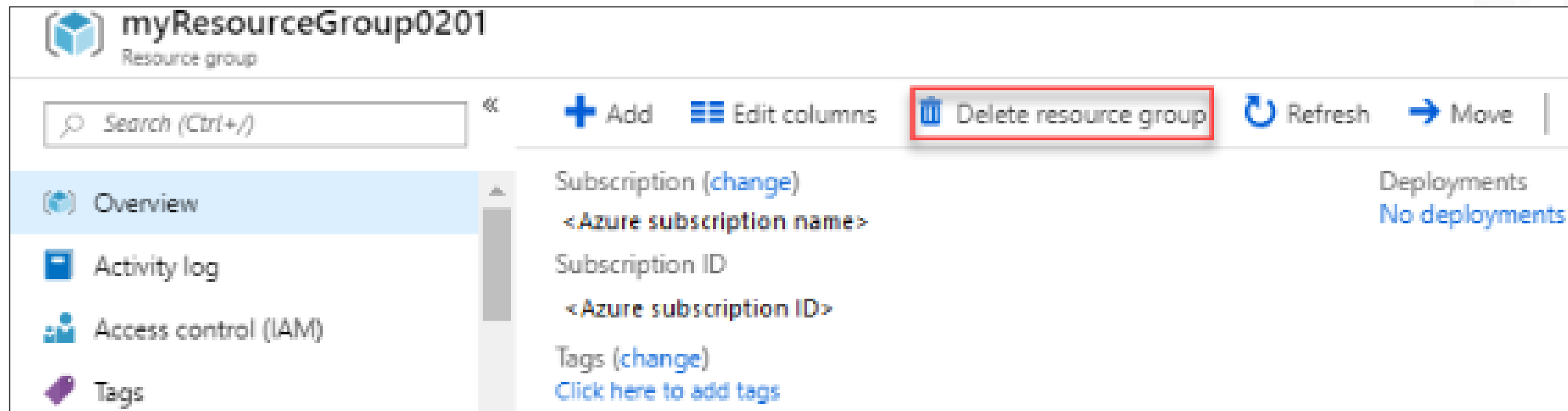
You can use each Microsoft Azure resource up to its quota. Each subscription has separate quotas and usage is tracked per subscription. If you reach a quota cap, you can request an increase via Help + Support. [Learn more](#) [Request Increase](#)

Quota	Provider	Location	Usage
Total Regional vCPUs	Microsoft.Compute	East US	25 % 25 of 100
Total Regional vCPUs	Microsoft.Compute	West Europe	21 % 21 of 100
Total Regional vCPUs	Microsoft.Compute	Central US	17 % 17 of 100
Standard Dv2 Family vCPUs	Microsoft.Compute	West Europe	16 % 16 of 100
Standard Dsv2 Family vCPUs	Microsoft.Compute	Central US	14 % 14 of 100

Source: <https://learn.microsoft.com/en-us/training/modules/use-azure-resource-manager/8-determine-resource-limits>

Remove Resources and Resource Groups

- Remove Azure resources that are no longer used, to prevent unexpected charges
- Remove individual resources or the resource group



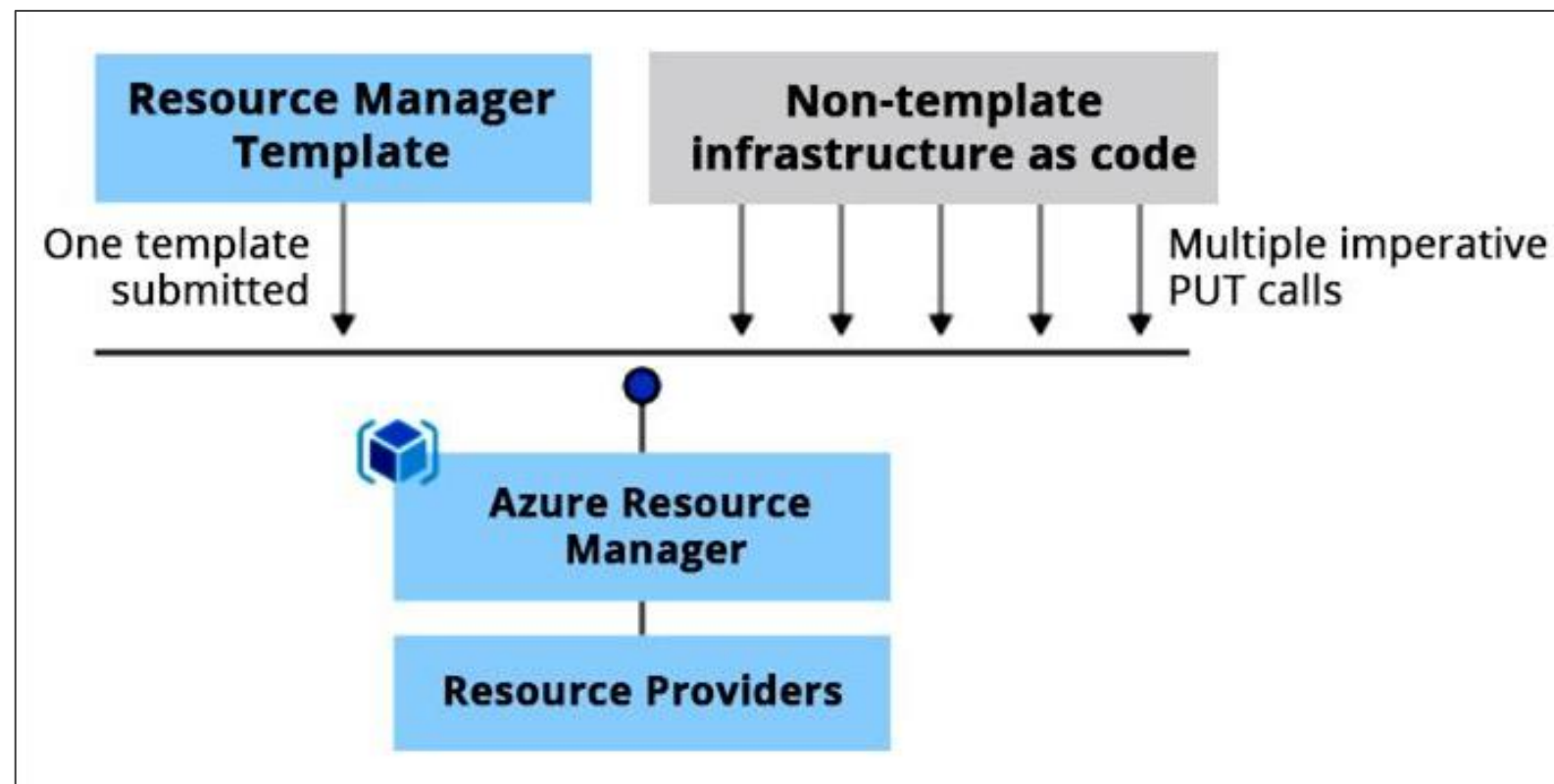
Source: <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/delete-resource-group?tabs=azure-portal>

ARM Templates

An Azure Resource Manager (ARM) template defines all the Resource Manager resources in a deployment.

Key Advantages

- Improves consistency
- Expresses complex deployments
- Reduces manual, error-prone tasks
- Expresses requirements through code
- Promotes reuse
- Is modular and can be linked



Source: <https://learn.microsoft.com/en-us/azure/azure-resource-manager/templates/overview>

ARM Template Schema

It presents the different sections of a template and the properties that are available in those sections.

```
{ "$schema":  
  "https://schema.management.azure.com/sch  
emas/2019-04-  
01/deploymentTemplate.json#",  
  "languageVersion": "", "contentVersion":  
  "", "apiProfile": "", "definitions": {  
  }, "parameters": { }, "variables": { },  
  "functions": [ ], "resources": [ ], /*  
or "resources": { } with languageVersion  
2.0 */ "outputs": { } }
```

- It is written in JSON.
- It is a collection of key-value pairs.
- Each key is a string.
- Each value can be a string, number, Boolean expression, list of values, or objects.

ARM Template Parameters

The template can be reused for different environments by providing different values for the parameters.

```
"parameters": {
  "adminUsername": {
    "type": "string",
    "metadata": {
      "description": "Username for the VM."
    }
  },
  "adminPassword": {
    "type": "securestring",
    "metadata": {
      "description": "Password for the VM."
    }
  }
}
```

Parameters specify which values are configurable when the template runs.

The example shown has two parameters:

- For a VM's username (adminUsername)
- For its password (adminPassword)

Quick Start Templates

The screenshot displays a grid of four Azure Quick Start Templates. Each template card includes a title, a date, a description, and buttons for 'Azure' and 'JSON' formats. An 'Add' button is located at the bottom right of each card.

Template Name	Date	Description	Formats
SAS 9.4 and Viya Quickstart Template for Azure	07/06/2022	The SAS® 9.4 and Viya QuickStart Template for Azure deploy these products on the cloud: SAS® Enterprise BI Server 9.4, SAS® Enterprise Miner 15.1, and SAS®...	Azure, JSON
App Service Quickstart - Windows app	07/06/2022	This template is to be used with http://docs.microsoft.com/azure/app-service/quickstart-arm-template	Azure, JSON
App Service Quickstart - Windows app	11/30/2022	This template is to be used with http://docs.microsoft.com/azure/app-service/quickstart-arm-template	Azure, JSON
Create a blob for the data factory copy data tool quickstart	10/11/2022	This template creates a blob storage and uploads a file for the copy data tool quickstart	Azure, JSON

- The Azure community provides the Resource Manager templates.
- These templates provide everything required to deploy solution or serve as a starting point for the template.

Source: <https://azure.microsoft.com/en-in/resources/templates/>

Quickstart ARM Template



Duration: 10 min.

Problem Statement:

You've been tasked to configure Azure resources with ARM templates in PowerShell.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Explore the Quickstart templates gallery
2. Explore template
3. Run templates with PowerShell



FULL STACK

Azure Bicep Files

Bicep

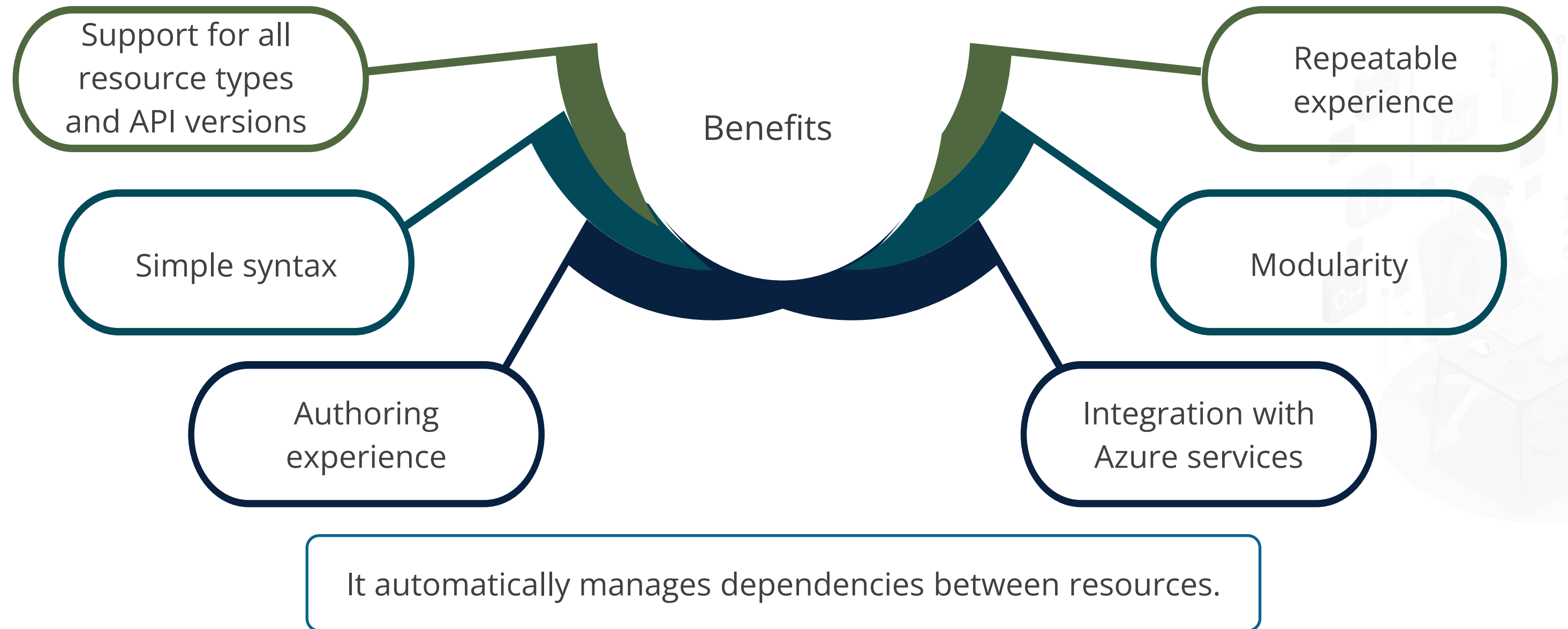
It is a domain-specific language (DSL) with a declarative syntax for deploying Azure resources.



- It defines the Azure infrastructure to be deployed.
- It can be used across the development cycle to repeatedly deploy the resources.
- It ensures consistent deployment.

Benefits of Bicep

It can be used instead of JSON to develop ARM templates.



Bicep Vs. ARM Template

Syntax complexity

- Bicep syntax is concise and developer friendly.
- It reduces verbosity and complexity in infrastructure code.

Developer experience

- It is designed to enhance the developer experience with more intuitive and readable syntax.

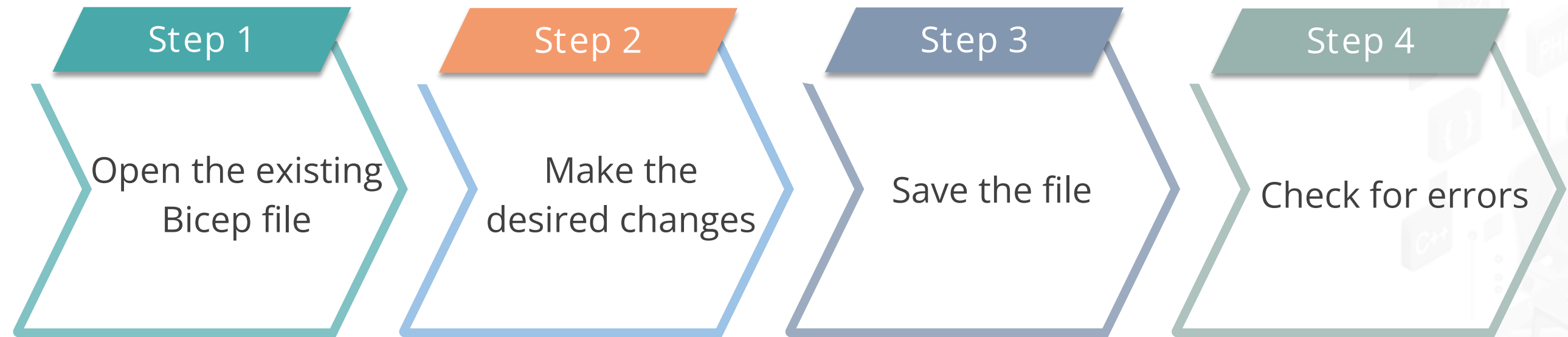
Abstraction

- Bicep is a transparent abstraction of ARM template JSON.

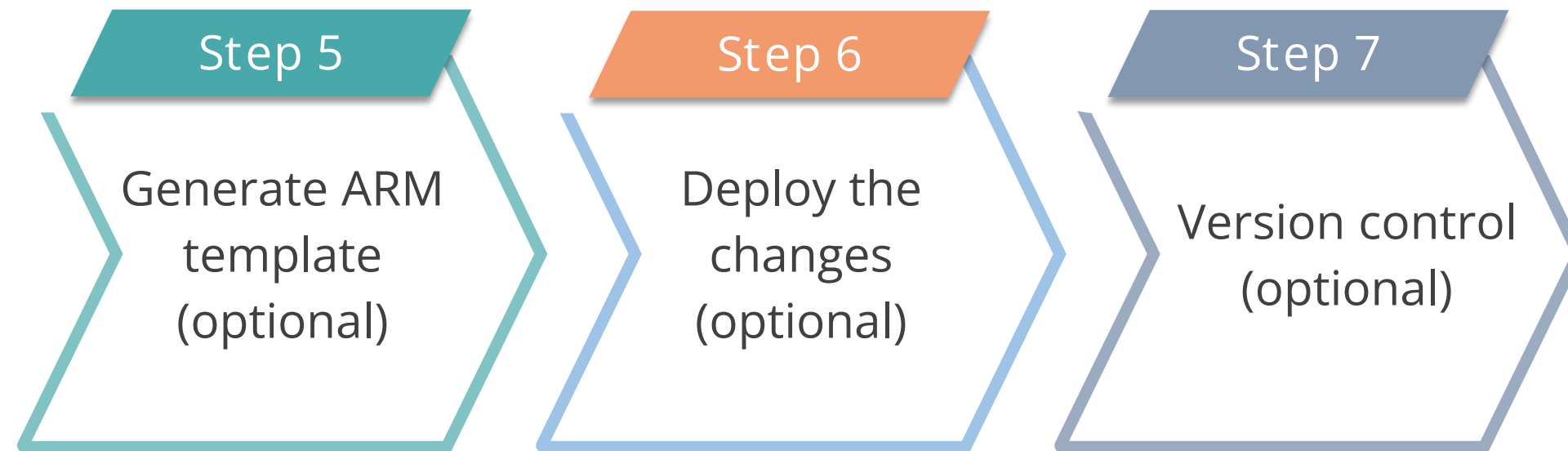
Conversion

- The Bicep CLI can convert Bicep files into ARM template JSON during deployment.

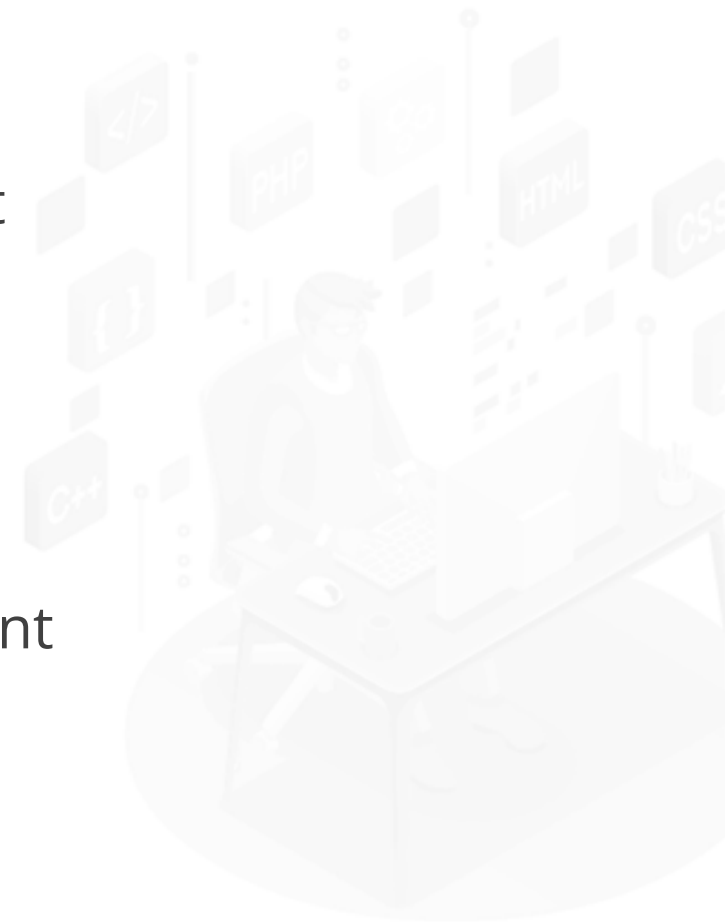
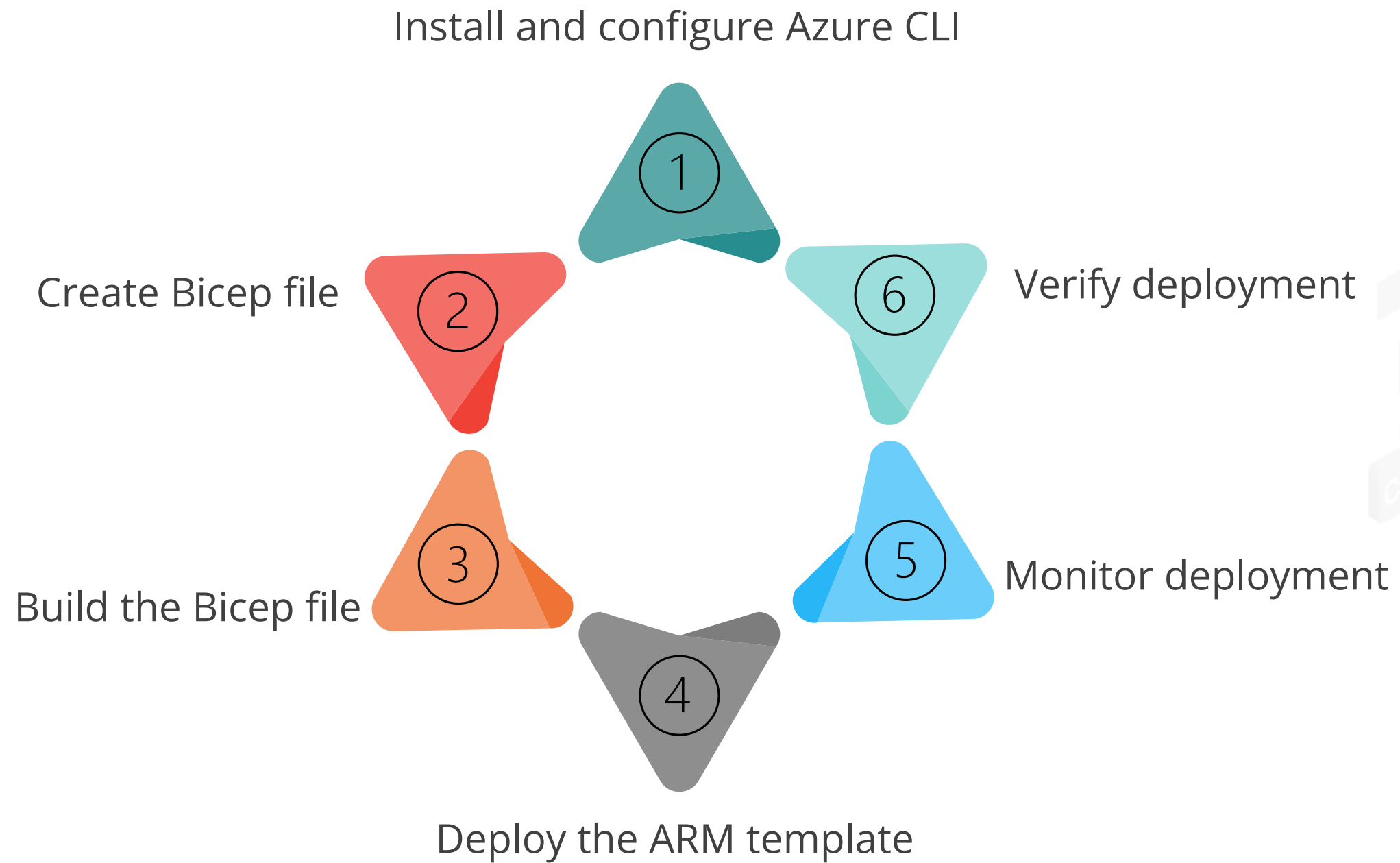
Modifying an Existing Bicep File



Modifying an Existing Bicep File



Deploy Resources Using Bicep File



Azure Cost Savings

Methods of cost saving:

- Azure Reservations: Help users save money by pre-paying for services
- Azure Hybrid Benefits: Employ Windows server and SQL server on-premises licenses with software assurance
- Azure Credits: Offers monthly credit benefit that allows one to experiment with, develop, and test new solutions on Azure
- Regions: Choose low-cost locations and regions
- Budgets: Help plan for and drive organizational accountability
- Pricing Calculator: Provides estimates in all areas of Azure, including compute, networking, storage, web, and databases

FULL STACK

Manage Role-Based Access Control (RBAC)

Create Management Groups

They are used when the user has several subscriptions and needs the following:

A level of scope
above subscriptions

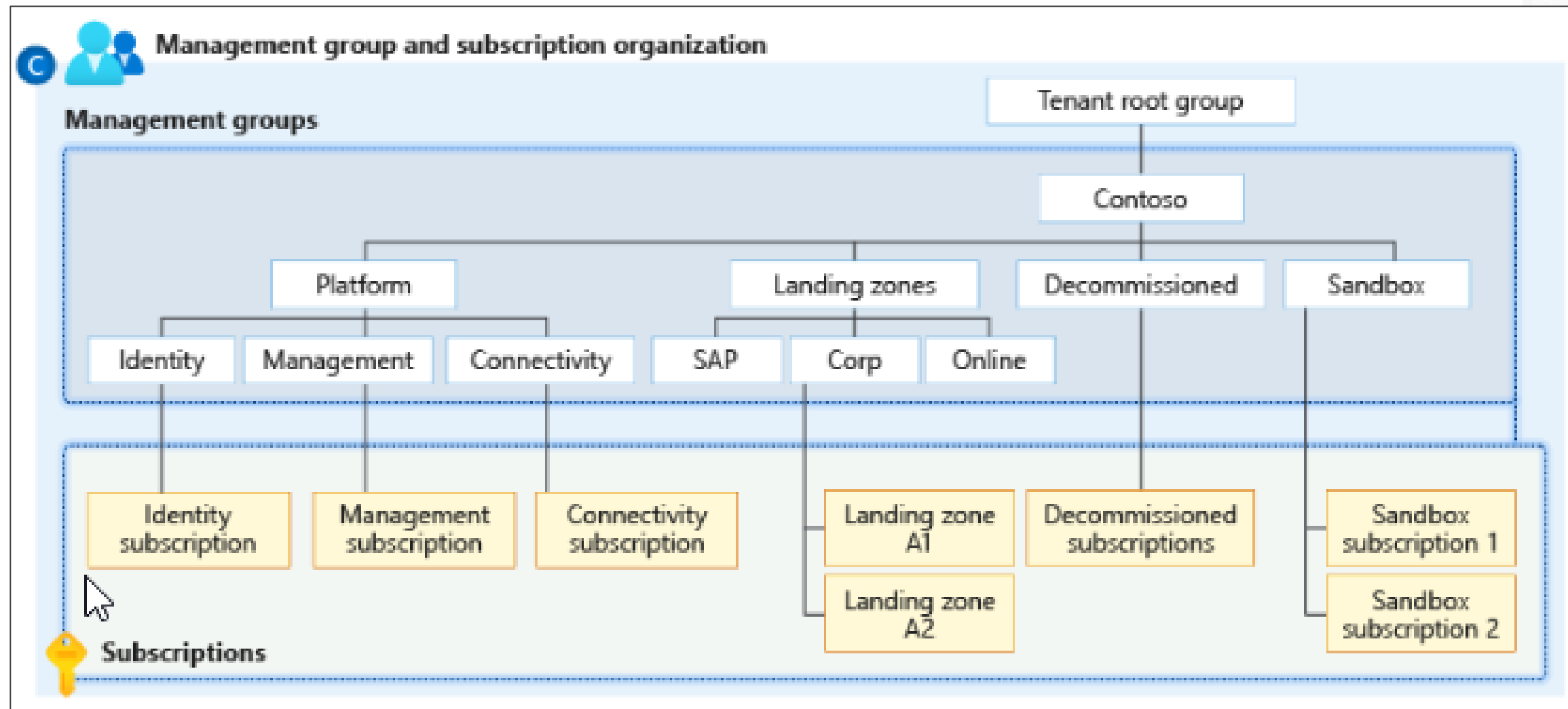
Help in targeting
policies and spending
budgets across
subscriptions and
inheritance down the
hierarchies

Compliance by the
organization
(business/teams)

Create Management Groups

It provides a flexible structure of management groups and subscriptions to organize the resources into a hierarchy for unified policy and access management.

Example: Creating a hierarchy for governance using management groups



Source: <https://learn.microsoft.com/en-us/azure/governance/management-groups/overview>

Creating Management Groups



Problem Statement:

You've been tasked to create management groups.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a management group



Configure Azure Policies

It is a service to create, assign, and manage policies that evaluate and scan for non-compliant resources.



Features

- Enforcement and compliance
- Application of policies at scale
- Bulk remediation

Configure Azure Policies

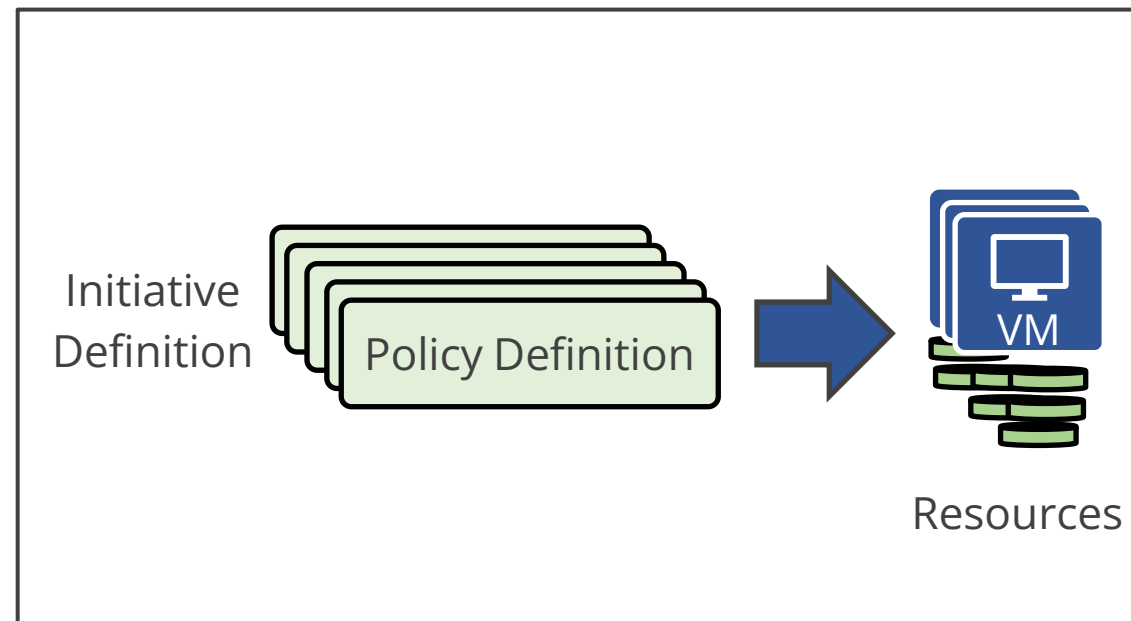
Use cases for implementing Azure policies are as follows:



- Allowed resource types
- Allowed virtual machine SKUs
- Allowed locations
- Required tag and its value
- Allowed Backup for virtual machines

Configure Azure Policies

To implement Azure policies, follow these steps:



- Create policy definitions
- Create initiative definitions
- Scope the initiative definition
- Determine compliance

1. Create Policy Definitions

There are many built-in policy definitions. The user can sort by category to choose any policy.

Policy definition

New Policy definition

BASICS

Definition location *

Visual Studio Enterprise Subscription ✓ ...

Name * ⓘ

Github Sample Policy ✓

Description

A Sample Policy from Github

Category ⓘ

☒ Create new ☐ Use existing

Category

POLICY RULE

↓ [Import sample policy definition from GitHub](#)

↗ [Learn more about policy definition structure](#)

The highlights are:

- Many policy definitions are available
- Users can import policies from GitHub
- Policy definitions have a specific JSON format
- Users can create custom policy definitions

2. Create Initiative Definitions

Once the user determines which policy definitions they need, they can create an initiative definition.

Initiative definition

New Initiative definition

Basics

Policies

Groups

Initiative parameters

Policy parameters

Review + create

An initiative definition is a collection of policy definitions that are tailored towards achieving a singular overarching goal. Initiative definitions simplify managing and assigning policy definitions by grouping them as a single assignable object.

Initiative location *

Visual Studio Enterprise Subscription

Name *

East Region

Description

East Region Initiative definition

Category

Create new

Use existing

General

Version

Review + create

Cancel

Previous

Next

Group policy definitions:

- Include one or more policies
- Require Planning

3. Scope the Initiative Definitions

Once the Initiative Definition is created, the user can assign the definition to a scope.

- The scope enforces the policy.
- The user can select the subscription, and optionally, the resource group.

The screenshot displays the 'Policy | Assignments' page in the Azure portal. The left sidebar contains navigation links: Overview, Getting started, Compliance, Remediation, Events, and an 'Authoring' section with 'Assignments' (selected), 'Definitions', and 'Exemptions'. The main content area has a search bar and buttons for 'Assign policy', 'Assign initiative', and 'Refresh'. Below these are filters for 'Scope' (set to 'Visual Studio Enterprise Subscri...'), 'Definition type' (set to 'All definition types'), and a search box. A purple banner message states: 'Now create custom non-compliance messages for policy assignments. Learn more https://aka.ms/policyassignmentnoncompliancmessage'. A summary row shows 'Total Assignments 1', 'Initiative Assignments 1', and 'Policy Assignments 0'. A table lists the assignments:

Assignment name ↑↓	Scope ↑↓	Type ↑↓
ASC Default (subscription: 35153c2e-cbb6-474f-a956-44c439f2a204)	Visual Studio Enterprise Subscription	Initiative

4. Determine Compliance

Use the Compliance blade to review:

- Non-compliant initiatives
- Non-compliant policies
- Non-compliant resources

The screenshot displays the Azure Policy | Compliance blade. The left sidebar contains navigation links: Overview, Getting started, Compliance (selected), Remediation, Events, and an Authoring section with Assignments, Definitions, and Exemptions. The main content area shows a search bar and action buttons: Assign policy, Assign initiative, and Refresh. Below these are filters for Scope (Visual Studio Enterprise S...), Type (All definition types), Compliance state (All compliance states), and a search box (Filter by name or ID...). The compliance summary includes: Overall resource compliance at 0% (0 out of 4); a donut chart for Resources by compliance state showing 0 Compliant, 0 Exempt, and 4 Non-compliant; Non-compliant initiatives at 1 out of 1; and Non-compliant policies at 32 out of 205. A table at the bottom lists the details for the 'ASC Default' policy.

Name	Scope	Compliance state	Resource compli...	Non-Compliant Resourc...	Non-compliant policies
ASC Default (subscription: 351...	Visual Studio Enterprise ...	Non-compliant	0% (0 out of 4)	4	32

Implementing an Azure Policy



Duration: 20 min.

Problem Statement:

You've been tasked to implement Azure policies to enforce compliance and regulatory settings.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Assign a policy
2. Create and assign initiative definition
3. Check for compliance



Implement Role-Based Access Control

It is an authorization system built on Azure Resource Manager that provides fine-grained access management of Azure resources.

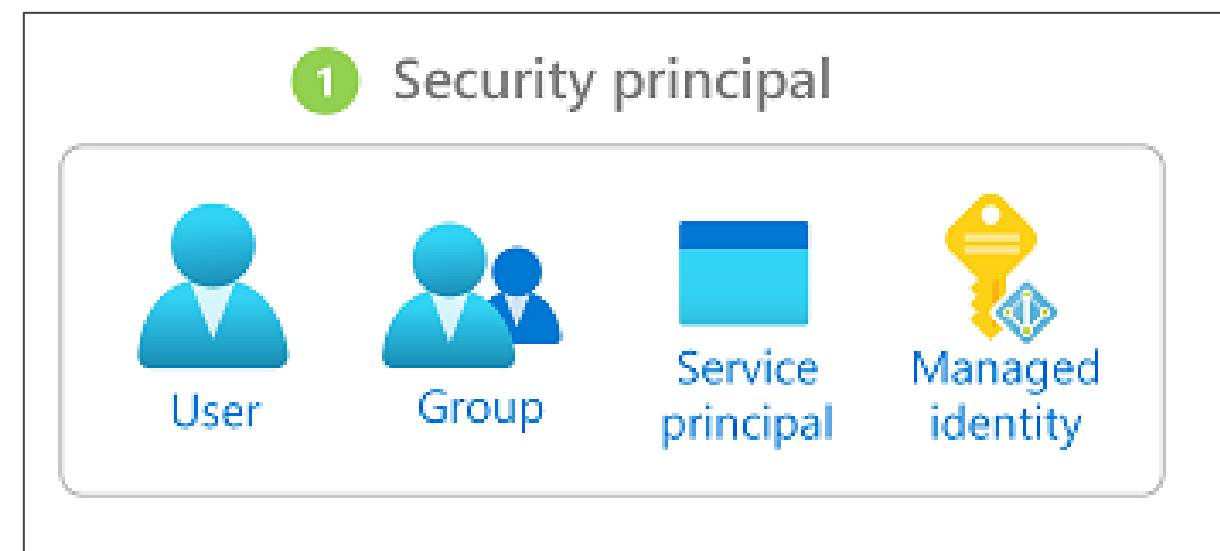
Azure RBAC enables the following:

- One user to manage virtual machines in a subscription and another user to manage virtual networks
- A DBA group to manage SQL databases in a subscription
- A user to manage all resources in a resource group, for example, virtual machines, websites, and subnets
- An application to access all resources in a resource group

Implement Role-Based Access Control

Concepts

- Security principal: Object that represents something that is requesting access to resources
- Role definition: Collection of permissions that lists the operations that can be performed
- Scope: Boundary for the level of access that is requested
- Assignment: Attaching a role definition to a security principal at a particular scope



Create a Role Definition

It is a collection of permissions listing the operations that can be performed. The following is an example of a definition:

Owner
Contributor
Reader
...
Backup Operator
Security Reader
User Access Administrator
Virtual Machine Contributor

Built-in

Reader Support Tickets
Virtual Machine Operator

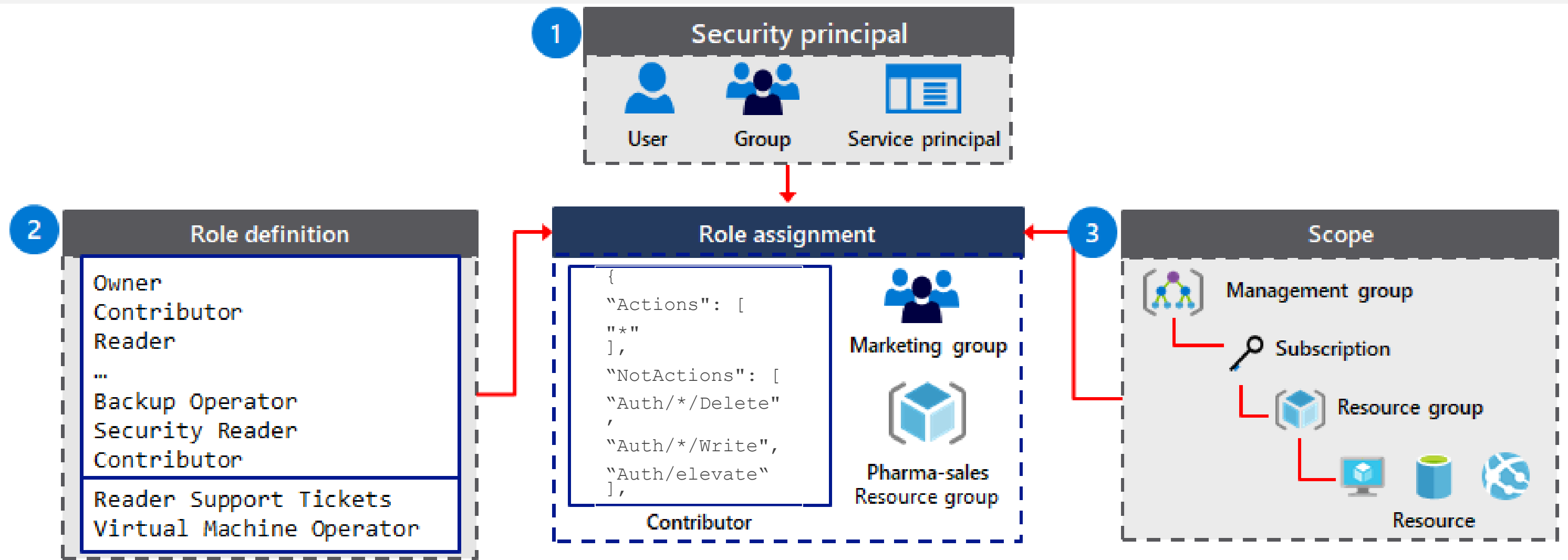
Custom

Contributor

```
{
  "actions": [
    "*"
  ],
  "notActions": [
    "Microsoft.Authorization/*/Delete",
    "Microsoft.Authorization/*/Write",
    "Microsoft.Authorization/elevateAccess/Action",
    "Microsoft.Blueprint/blueprintAssignments/write",
    "Microsoft.Blueprint/blueprintAssignments/delete",
    "Microsoft.Compute/galleries/share/action"
  ],
  "dataActions": [],
  "notDataActions": []
}
```

Create a Role Assignment

It is the process of binding a role definition to a user, group, or service principal at a scope for the purpose of granting access. The example below depicts the creation of a role assignment.



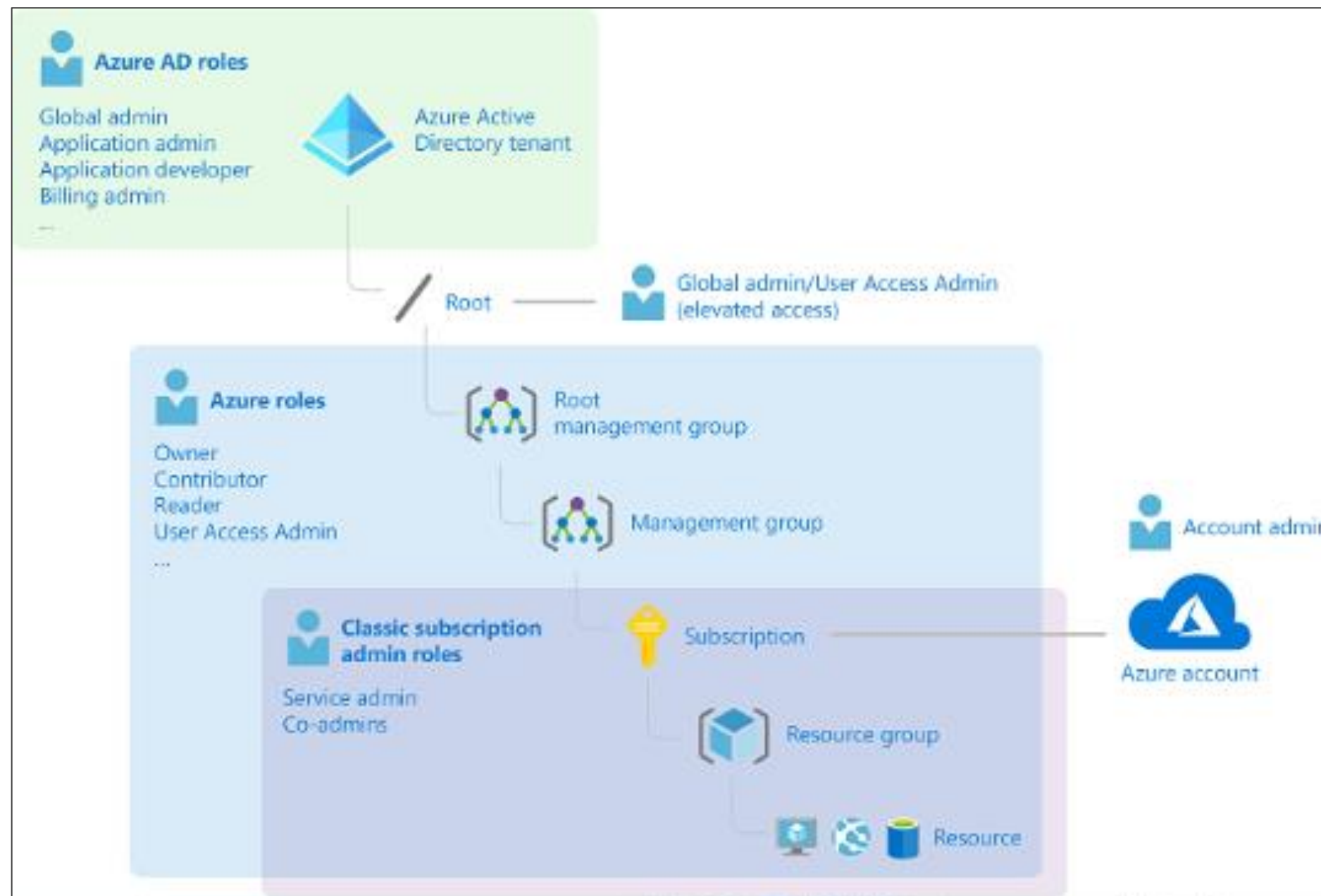
Source: <https://learn.microsoft.com/en-us/azure/role-based-access-control/overview>

Compare Azure RBAC Roles to Entra ID Roles

Azure RBAC roles	Entra ID roles
Manage access to Azure resources	Manage access to Entra ID objects
Scope can be specified at multiple levels	Scope is at the tenant level
Role information can be accessed in the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager templates, REST API	Role information can be found in Azure portal, Microsoft 365 admin portal, Microsoft Graph, Microsoft Entra ID PowerShell for Graph

Apply RBAC Authentication

The following image illustrates the differences in roles between Microsoft Entra ID (Azure AD) Admin and Azure RBAC:



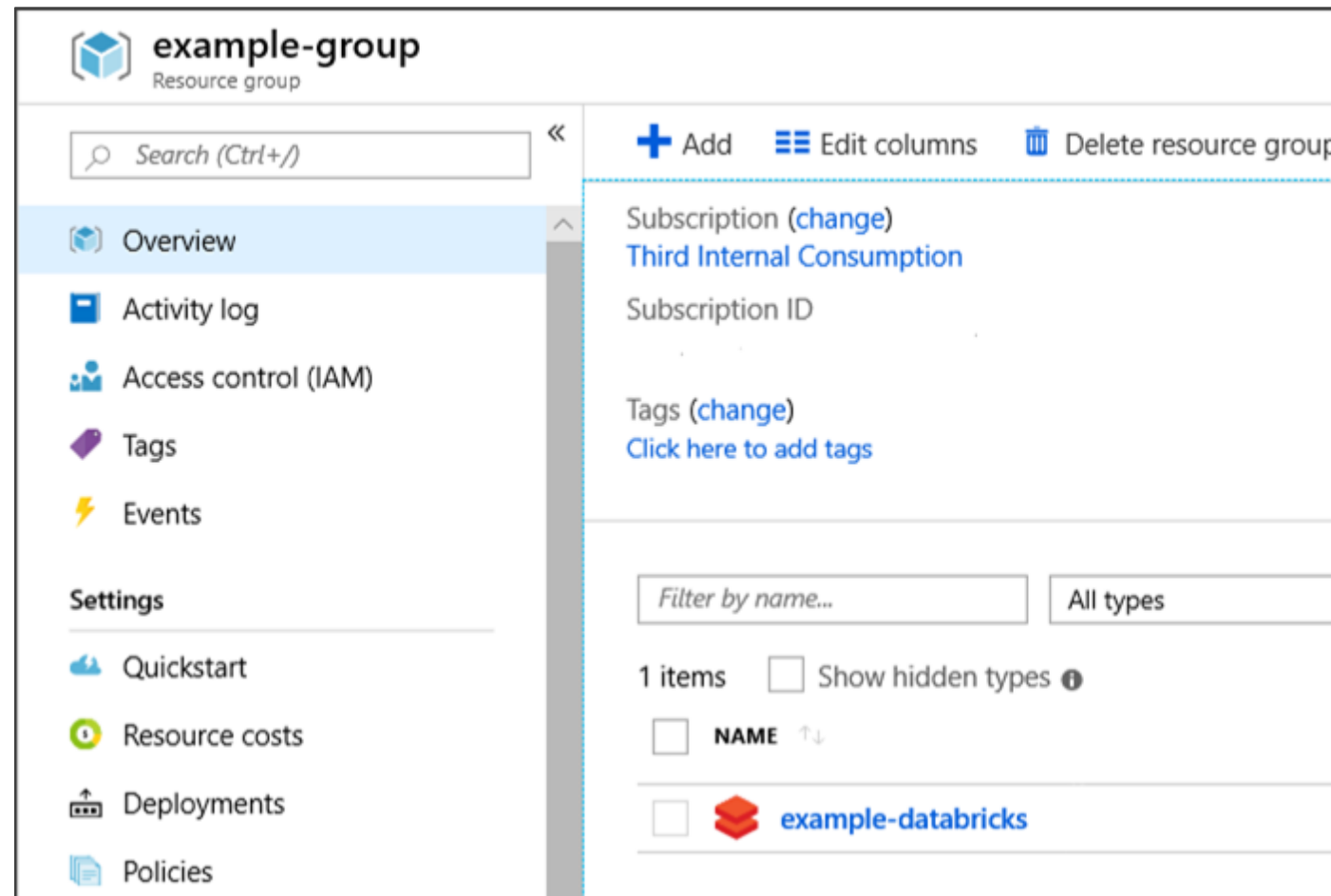
Source: <https://learn.microsoft.com/en-us/azure/role-based-access-control/rbac-and-directory-admin-roles>

Determine Azure RBAC Roles

RBAC Roles and Permissions in Azure:

RBAC role in Azure	Permissions	Notes
Owner	Has full access to all resources and can delegate access to others	The Service Administrator and Co-Administrators are assigned the Owner role at the subscription scope. This applies to all resource types
Contributor	Creates and manages all types of Azure resources but cannot grant access to others	This applies to all resource types
Reader	Views Azure resources	This applies to all resource types
User Access Administrator	Manages user access to Azure resources	This applies to managing access, rather than to managing resources

Resource Manager Locks



Features of locks:

- Locks can be associated with a subscription, resource group, or resource
- Locks are inherited by child resources

Types of locks:

- Read-only locks prevent any changes to the resource
- Delete locks prevent deletion

Creating Resource Lock



Duration: 10 min.

Problem Statement:

You've been tasked to create a resource lock to have more control access.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

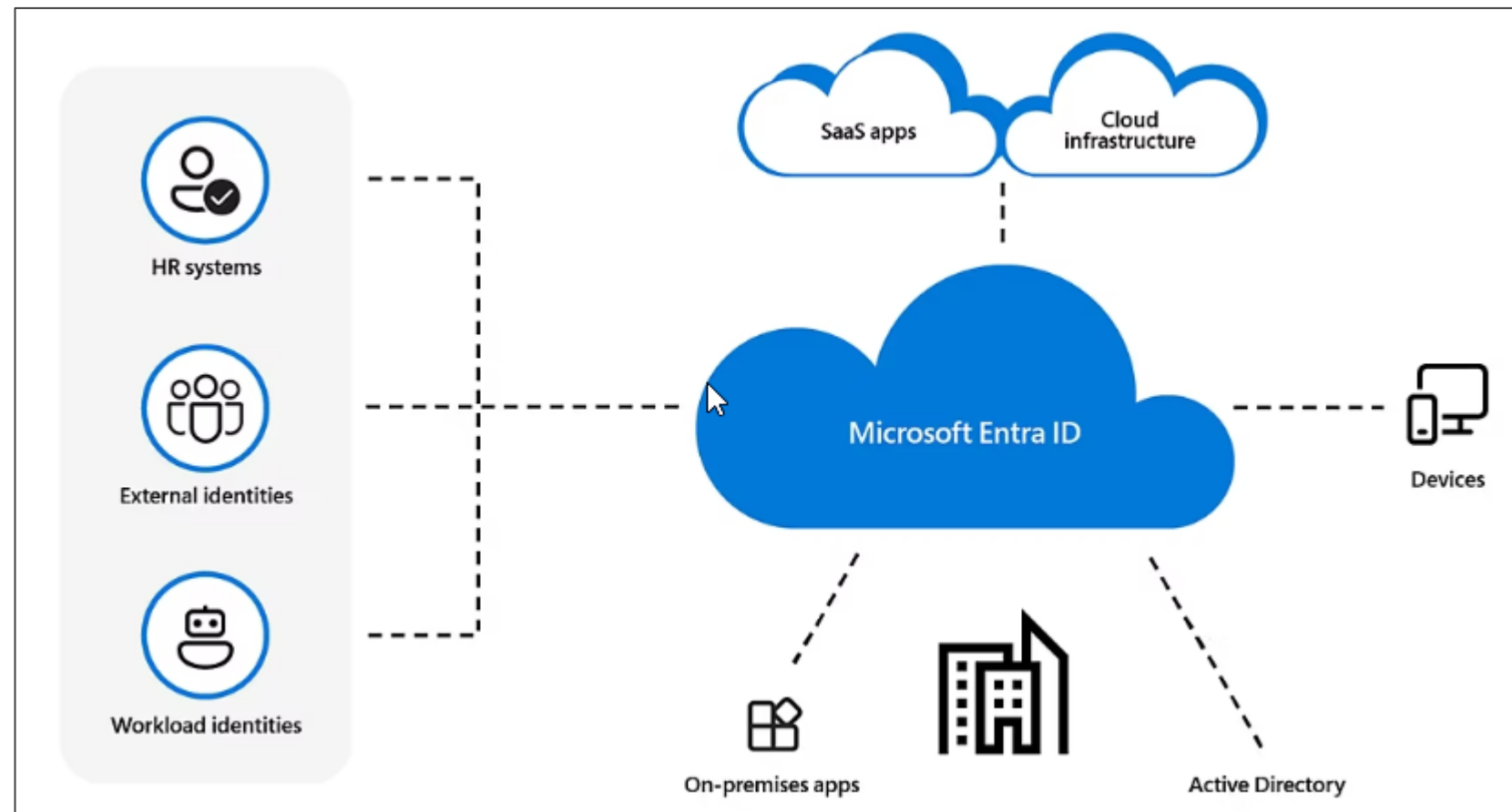
1. Configure locks on Azure resources



FULL STACK

Administer Identity

Microsoft Entra ID



- It is a cloud-based suite of identity management capabilities that enable securely managed access to Azure services and resources for the users.
- It provides application management, authentication, device management, and hybrid identity.

Source: <https://www.microsoft.com/en-in/security/business/identity-access/microsoft-entra-id>

Entra ID Concepts

Concept	Description
Identity	It is an object that can be authenticated.
Account	It is an identity that has data associated with it.
Microsoft Entra account	An identity created through Microsoft Entra ID or another Microsoft cloud service, such as Microsoft 365. Identities are stored in Microsoft Entra ID and accessible to your organization's cloud service subscriptions. This account is also sometimes called a Work or school account.
Azure tenant	It is a dedicated and trusted instance of Microsoft Entra ID.
Microsoft Entra directory	Each Azure tenant has a dedicated and trusted Microsoft Entra directory.

Entra ID Concepts



Microsoft Entra ID (Azure AD)

- It is an identity solution designed for HTTP and HTTPS communications.
- It can be queried using REST API over HTTP and HTTPS, instead of LDAP.
- It uses HTTP and HTTPS protocols, such as SAML, WS-Federation, and OpenID Connect, for authentication (and OAuth for authorization) instead of Kerberos.
- It includes federation services and many third-party services (such as Facebook).
- It's users and groups are created in a flat structure, and there are no Organizational Units (OUs) or Group Policy Objects (GPOs).

Compare Entra DS to Entra ID

The following table depicts the difference between Entra ID and Entra DS.

Microsoft Entra ID	Microsoft Entra Domain Services
Primarily an identity solution for cloud-based Applications	An identity solution for on-prem application
Queried over HTTP and HTTPS protocol	Queried over LDAP protocol
Includes federation and third-party services (like Twitter, LinkedIn)	Does not include any federated or third-party providers
OAuth or OpenID connect-based protocols	Kerberos and NTLM protocols
Resources are organized in a flat structure	The resources are divided into Organizational units or Group Policy objects

Microsoft Entra ID Editions

Microsoft Entra ID comes in four editions: Free, Office 365 apps, Premium P1, and Premium P2.

Feature	Microsoft Entra ID Free	Microsoft Entra ID P1	Microsoft Entra ID P2	Microsoft Entra ID Governance
Authentication, Single sign-on, and application access	X	X	X	
Administration and hybrid identity	X	X	X	
End user self service	X	X	X	X
Multifactor authentication and conditional access	X	X	X	
Identity protection			X	
Event logging and reporting	X	X	X	
Identity governance	X	X	X	X

Implement Self-Service Password Reset

By enabling SSPR, users can:

Determine who can use a self-service password reset

Choose the number of authentication methods required and the methods available

Register for SSPR (same process as MFA)

Implement Self-Service Password Reset

The image below shows the password reset properties:

Password reset | Properties

Default Directory - Azure Active Directory

<< Save Discard

Diagnose and solve problems

Manage

- Properties
- Authentication methods
- Registration
- Notifications
- Customization
- On-premises integration
- Administrator Policy

Activity

- Audit logs
- Usage & insights

Troubleshooting + Support

- New support request

Self service password reset enabled ⓘ

None Selected All

Select group ⓘ

AAD DC Administrators

i [These settings only apply to end users in your organization. Admins are always enabled for self-service password reset and are required to use two authentication methods to reset their password. Click here to learn more about administrator password policies.](#)



Implementing Self-Service Password Reset (SSPR)



Problem statement

You have been tasked to implement a self-service password reset.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a test user in Microsoft Entra ID
2. Configure SSPR for the test user
3. Reset SSPR for the test user



Configure User and Group Accounts

Create User Accounts

Highlights:

- All users must have an account
- The account is used for authentication and authorization
- Each user account has additional properties

Home > Simplilearn Solutions Pvt Ltd | Users >

Users

Search

+ New user ▾ Download users Bulk operations ▾ Refresh Manage view ▾ Delete Per-user MFA Got feedback?

All users Azure Active Directory is becoming Microsoft Entra ID. ⓘ

Audit logs

Sign-in logs

Diagnose and solve problems

Manage

Deleted users

Password reset

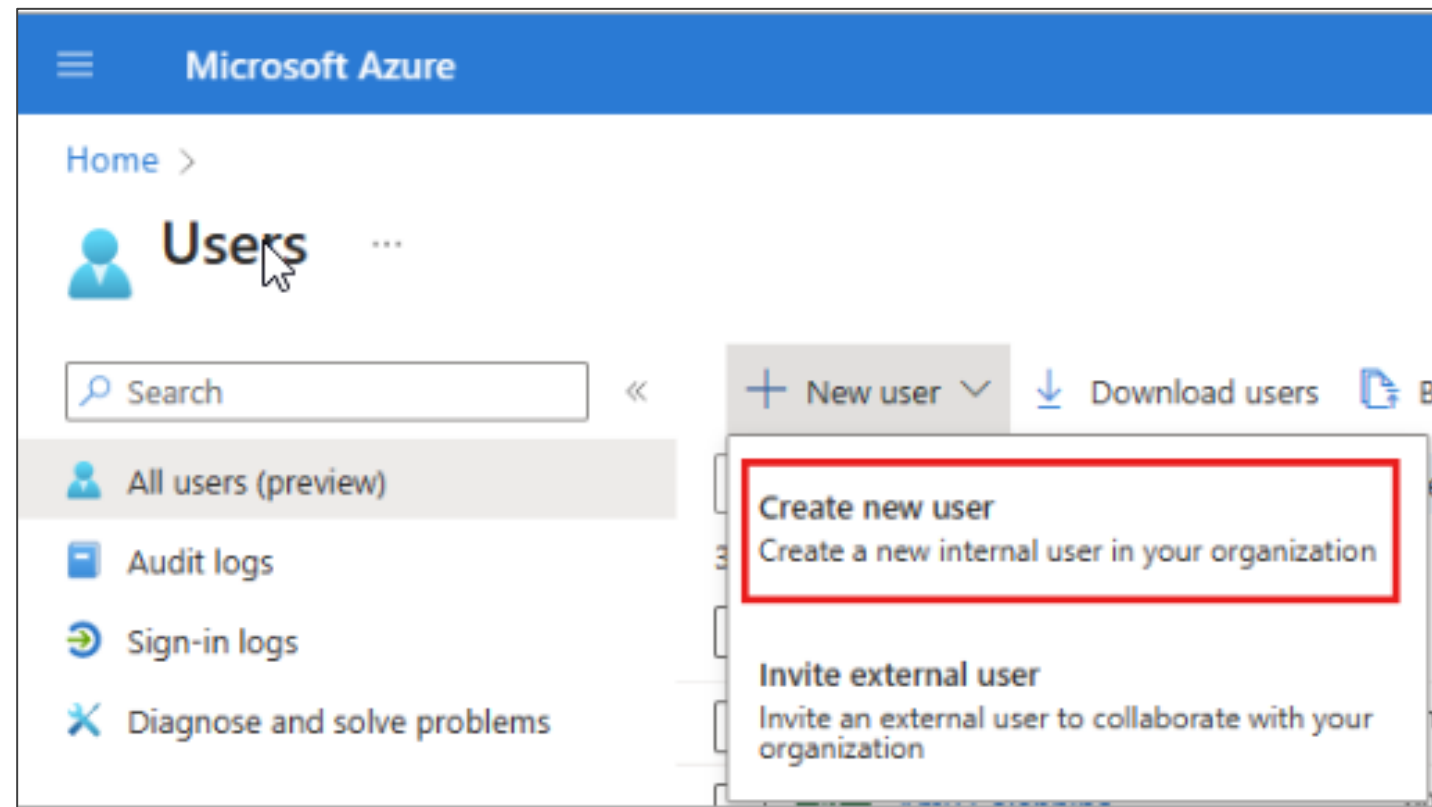
User settings

Search Add filter

3 users found

☐	Display name ↑	User principal name ↑↓	User type	On-premises sy...	Identities	Company name
☐	 Gowtham S	gowtham.s@simplilearn.c...	 Member	No	simplilearnsolution.onmicrosoft.cc	
☐	 gowtham.s	gowtham.s_simplilearn.ne...	 Guest	No	mail	
☐	 Guwalani Kunal	guwalani.kunal@simplilea...	 Member	No	simplilearnsolution.onmicrosoft.cc	
☐	 Hari Krishna	hari.krishna@simplilearn.c...	 Member	No	simplilearnsolution.onmicrosoft.cc	

Manage User Accounts



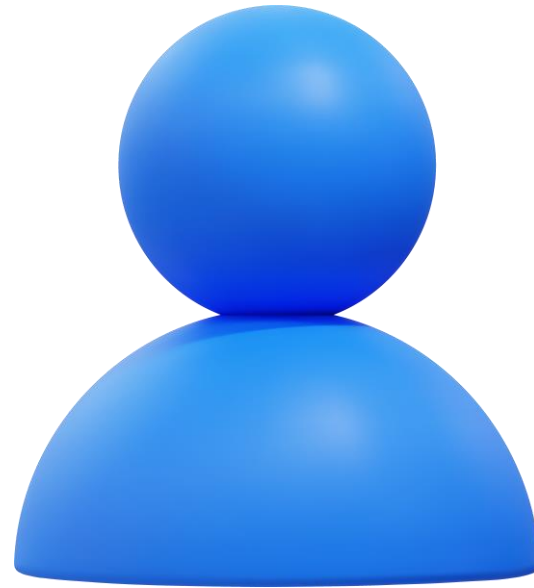
Things to consider when managing users:

- Must be a global administrator or a user administrator to manage users
- User profile (picture, job, contact info) is optional
- Deleted users can be restored within 30 days
- Sign-in and audit log information is available

Source: <https://learn.microsoft.com/en-us/azure/active-directory/fundamentals/add-users>

Manage External Users

Microsoft Entra External ID enables secure interactions with external users.



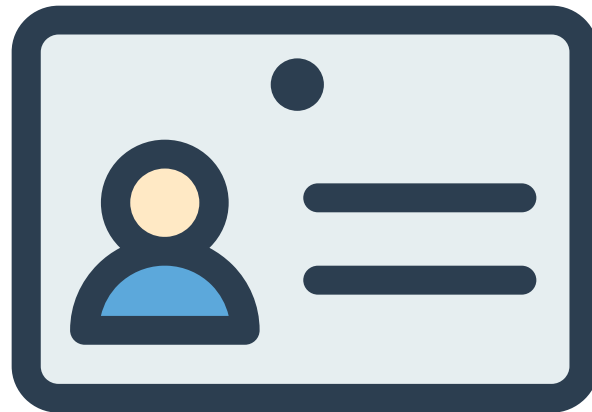
The following capabilities make up External Identities:

- B2B collaboration
- B2B direct connect
- Microsoft Entra ID B2C
- Microsoft Entra multi-tenant organization

With External Identities, external users can bring their own identities.

Manage Licenses in Entra ID

Only users with active licenses can access and use Microsoft Entra services, which are applied to their respective tenant and are non-transferable.



Available licence plans:

- Microsoft Entra ID free
- Microsoft Entra ID P1
- Microsoft Entra ID P2

Create Group Accounts

It allows to define two types of groups and offers three membership types.

Group types:

- Security
- Microsoft 365 groups

Membership types:

- Assigned
- Dynamic User
- Dynamic Device (Security groups only)

☐	Name ↑	Object Id	Group type	Membership type
☐	 online@simplilearn.com	00089ab7-bb2a-43a5-be8d-0904073ae675	Security	Assigned
	 opensesame-support@simplilearn.com	71d55901-52ce-4b02-9795-02a43dd9ab3d	Distribution	Assigned
	 operation_finance	89ba958e-4e55-4f5c-af9c-d6e5b2dc955f	Distribution	Assigned
☐	 Ops Managers	4c08273f-5eef-4a82-b41b-e73692c69a27	Microsoft 365	Assigned

Creating User and Group Accounts



Duration: 10 min.

Problem Statement:

You have been tasked to create user and group accounts to add the users to Azure subscriptions.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Add a new user
2. Create a new group and add members



Inserting Guest Users into the Directory



Duration: 10 min.

Problem Statement:

You have been tasked to add a guest user into the directory for inviting external (guest) users to the Azure subscription.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Add a guest user to the directory



Key Takeaways

- An ARM template is an IAC tool to create and version Azure Services.
- Azure Resource Manager is the single interface for managing all Azure resources.
- Azure resources can be managed by various tools like Portal, CLI, and PowerShell.
- Azure RBAC can be used to secure access to Azure services.



Key Takeaways

- Azure Policy can be used to meet regulatory and compliance standards for organizations.
- Azure cost management is the unified dashboard to manage all the cost and billing-related tasks.
- Azure Entra ID is used to manage all user and group creations.



Implementing Azure Directory and Assign Permissions



Project agenda: To implement Microsoft Entra ID, add and remove users from Microsoft Entra ID, and implement RBAC.

Description: You have been given a project to create Microsoft Entra ID users in the subscriptions. Once the users have been created, you need to assign permissions to these users.

Perform the following:

Create users in Active Directory and assign the application administrator permission to the newly created users.